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P. Puttonen (2016): Lecture on Short-Circuits, part of *Energy Transmission* lecture series, 8th-10th November 2016, FH Vorarlberg, Dornbirn

Short circuits

Energy Transmission
2026-04-13

Content

- ♦ **Introduction**
 - ♦ **Definition of short circuit, different types of short circuits, characteristics of the current**
- ♦ Short-circuit impedances of electrical equipment
 - ♦ Equivalent circuits, calculation of resistance and reactance of each component, examples, exercises
- ♦ Calculation of short circuit currents
 - ♦ Calculation assumptions, formulas, examples, exercises

Short-circuit

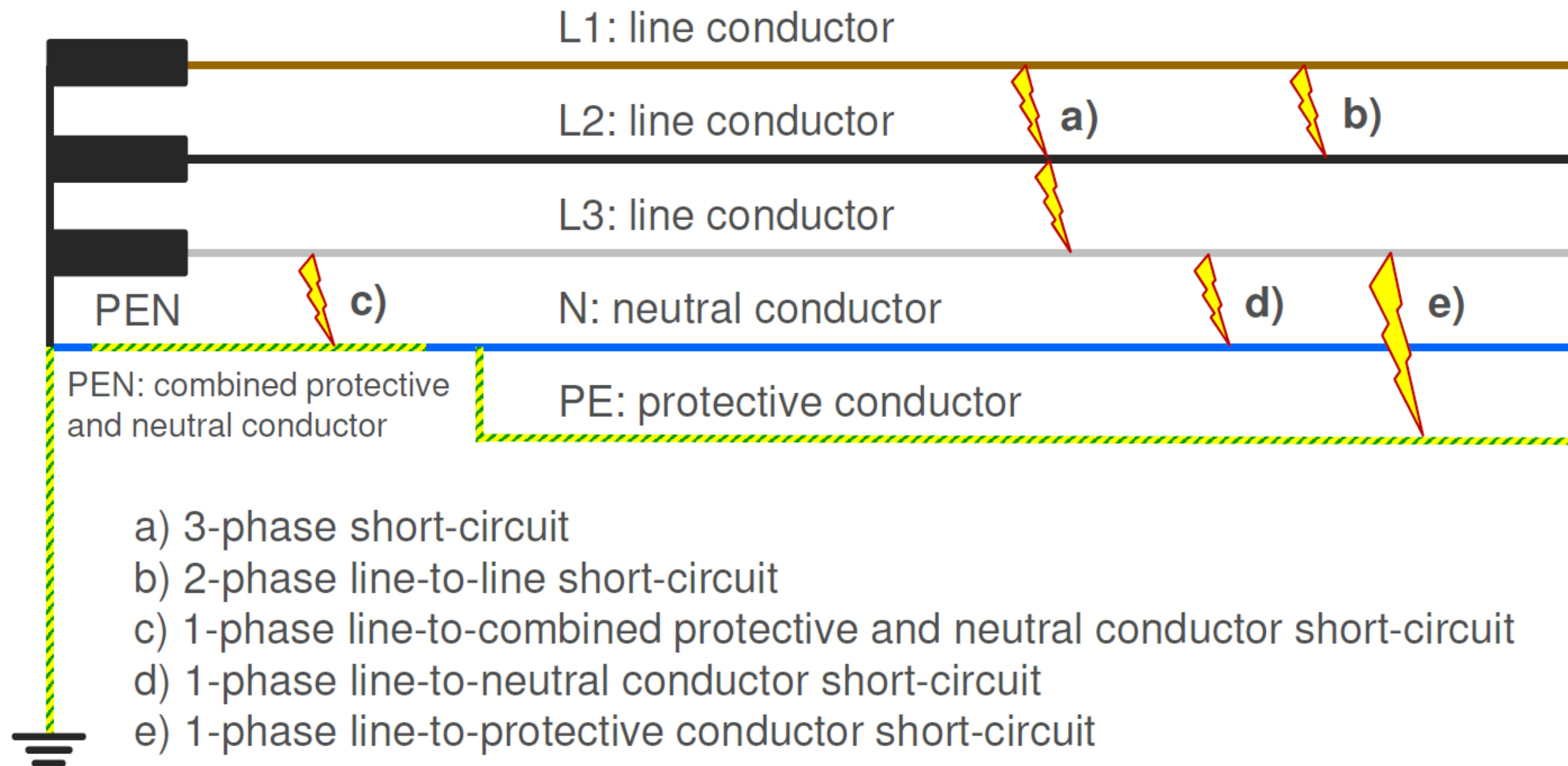
- ♦ A short-circuit is an accidental connection between conductors by a zero (solid short-circuit: busbar or wire) or non-zero impedance (impedant short-circuit: animals, insulation damages)



Short-circuit

- ◆ Short-circuit current is “over-current resulting from a short-circuit in an electric system”
- ◆ Multi-phase short-circuit: conductive path is between two or more line conductors
- ◆ Single-phase short-circuit: conductive path is between a line conductor and local earth

Short-circuit in a low-voltage system

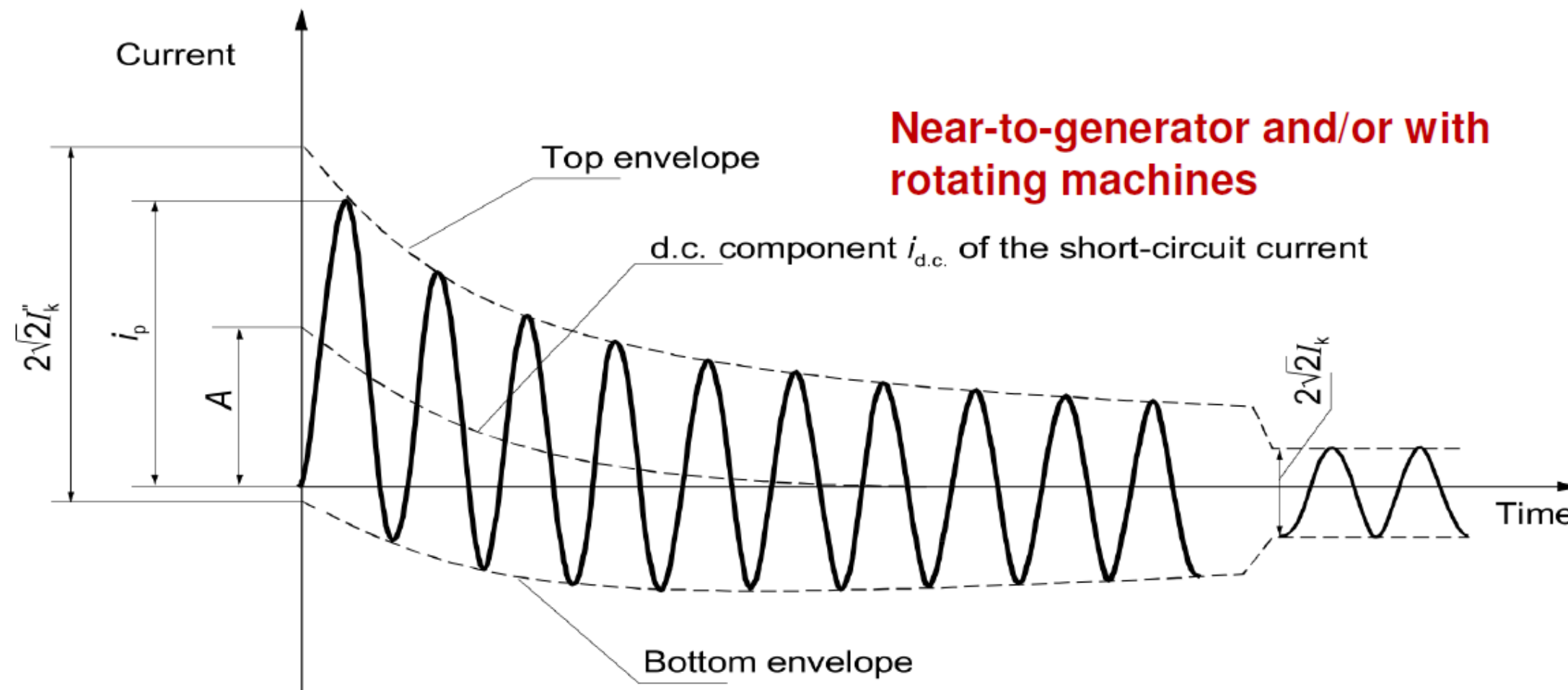


Short-circuit

- ♦ Short-circuit current I_{SC} is much higher than the rated load current I_e



Characteristics of short-circuit currents



I_k'' = initial symmetrical short-circuit current

i_p = peak short-circuit current

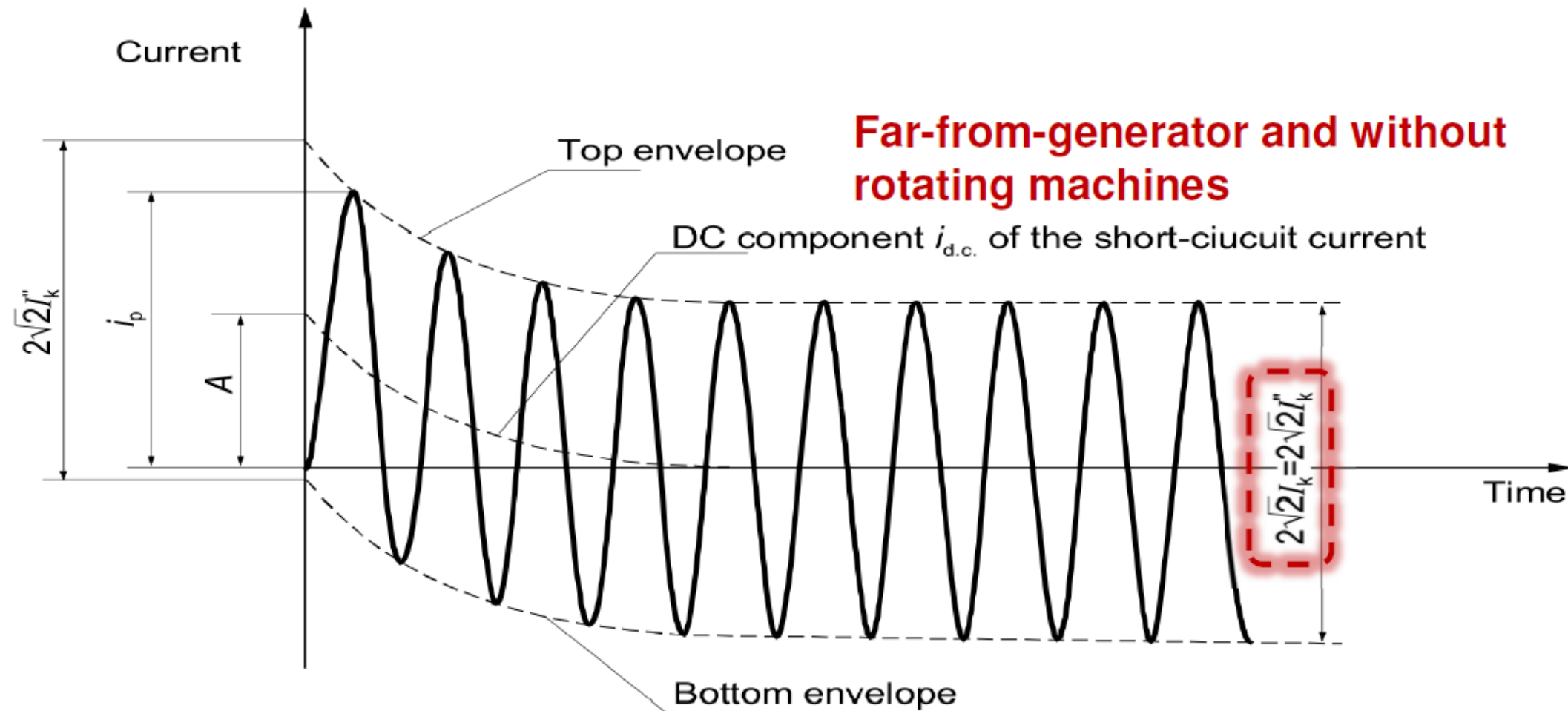
I_k = steady-state short-circuit current

$i_{d.c.}$ = d.c. component of short-circuit current

A = initial value of the d.c. component $i_{d.c.}$

This is the envelope of one phase, the envelope is on different level on other phases.

Characteristics of short-circuit currents



I_k'' = initial symmetrical short-circuit current

i_p = peak short-circuit current

I_k = steady-state short-circuit current

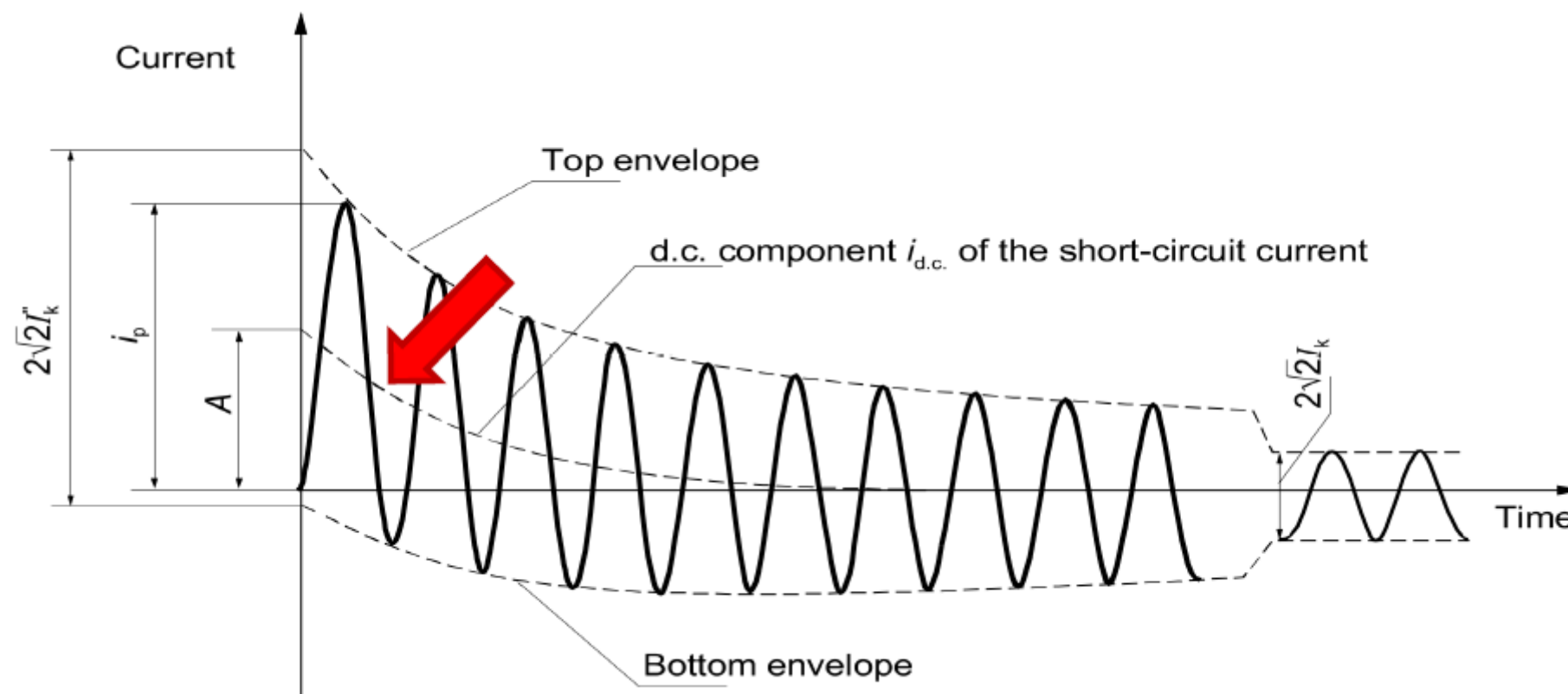
$i_{d.c.}$ = d.c. component of short-circuit current

A = initial value of the d.c. component $i_{d.c.}$

Now there is no decaying a.c. component

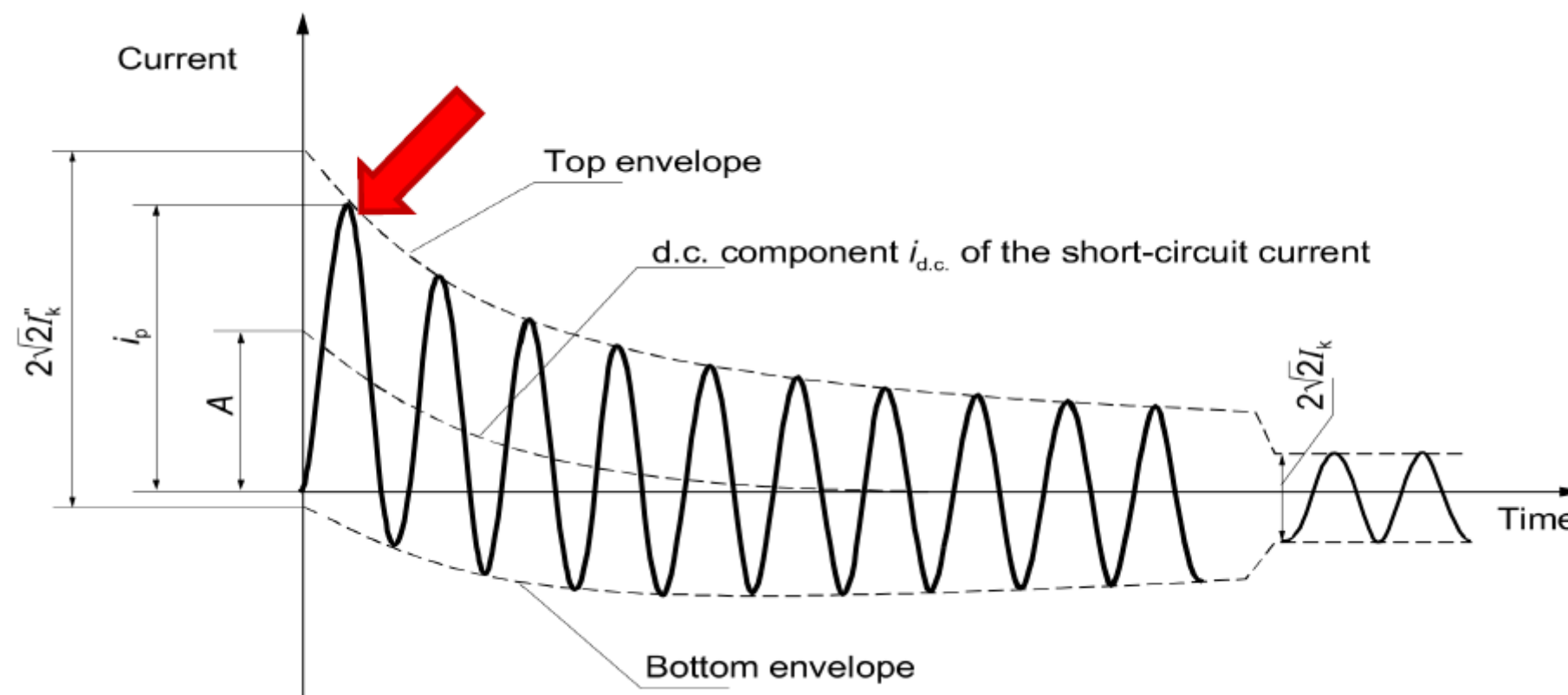
Characteristics of short-circuit currents

- ♦ Time constant of the decaying d.c. component depends on the R/X-ratio of the network
 - R/X-ratio depends on equivalent resistances and reactances of network components
 - Cables, transformers, etc.



Characteristics of short-circuit currents

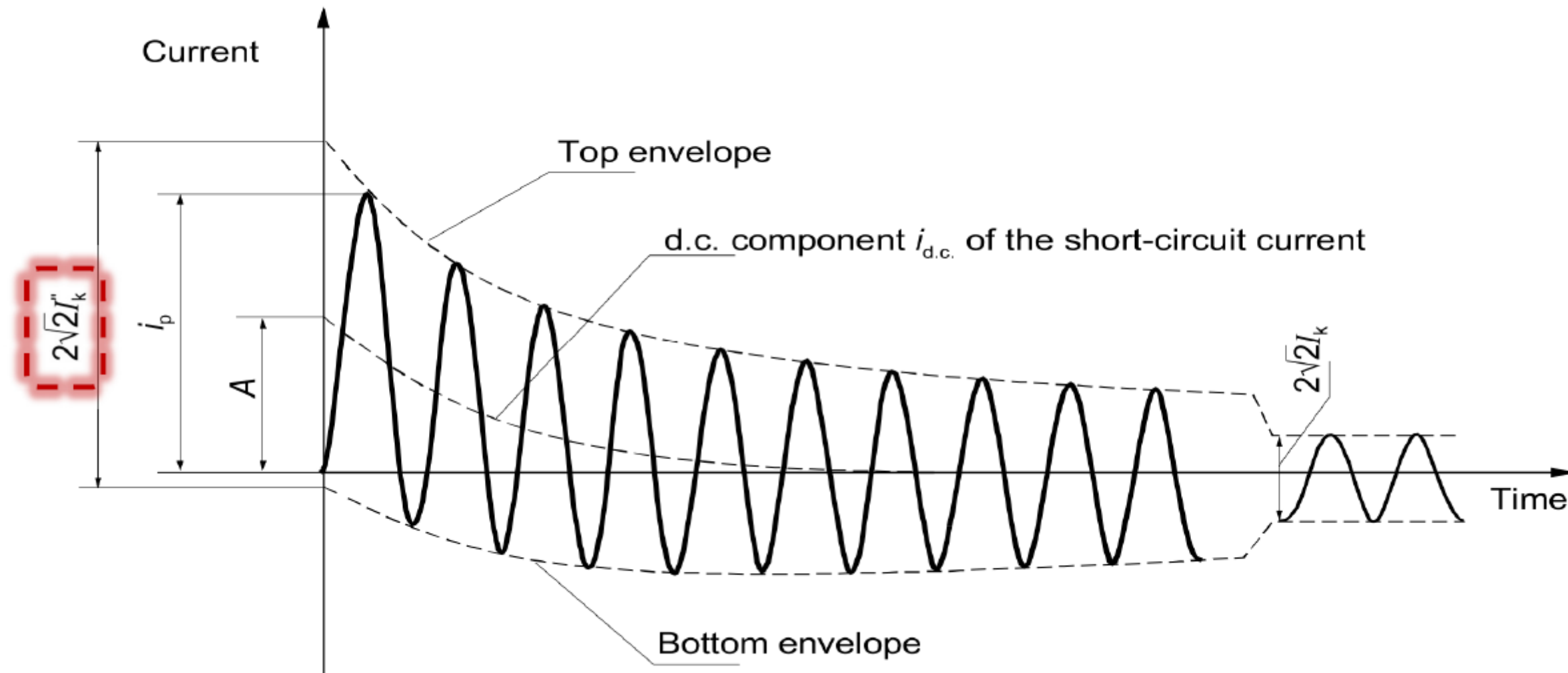
- ♦ Time constant of the decaying a.c. component depends on the properties of the rotating machines
 - Synchronous machines excitation
 - Group of asynchronous motors (total power of motors)



Characteristics of short-circuit currents

- ◆ Symmetrical short-circuit current
 - R.m.s. value of the a.c. symmetrical prospective shortcircuit current when the d.c. component of current is neglected
- ◆ Prospective short-circuit current
 - Current that would flow if the short-circuit is replaced by an ideal connection of negligible impedance

Characteristics of short-circuit currents

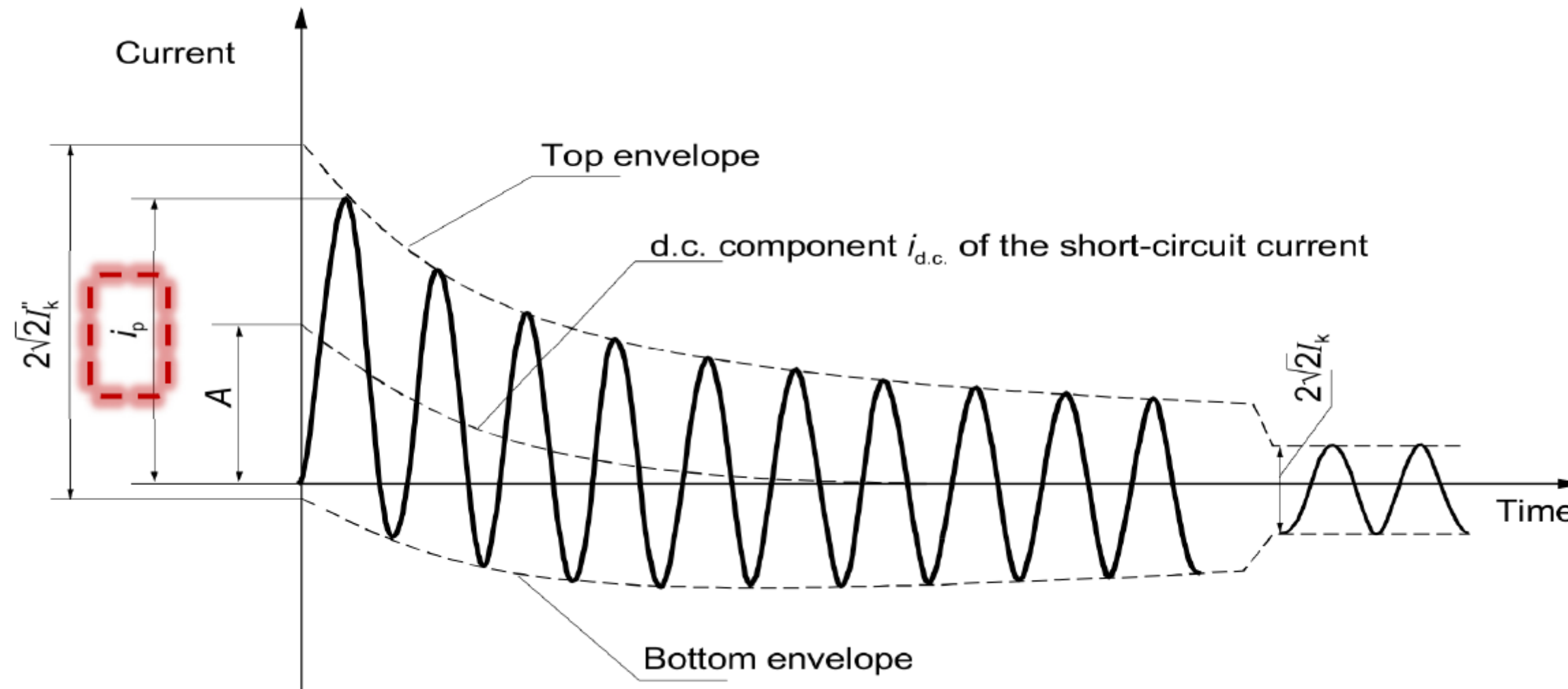


- ◆ Initial symmetrical short-circuit current I_k''
 - R.m.s. value of the a.c. symmetrical prospective short-circuit current at the beginning of short-circuit

Initial symmetrical s.c.-current I''_K

- ◆ Affects on the thermal stress of electrical equipment
- ◆ I''_K value is needed to ensure the protection equipment (e.g. circuit breaker) breaking capacity (current should be broken as soon as possible at the beginning of the short-circuit)

Characteristics of short-circuit currents

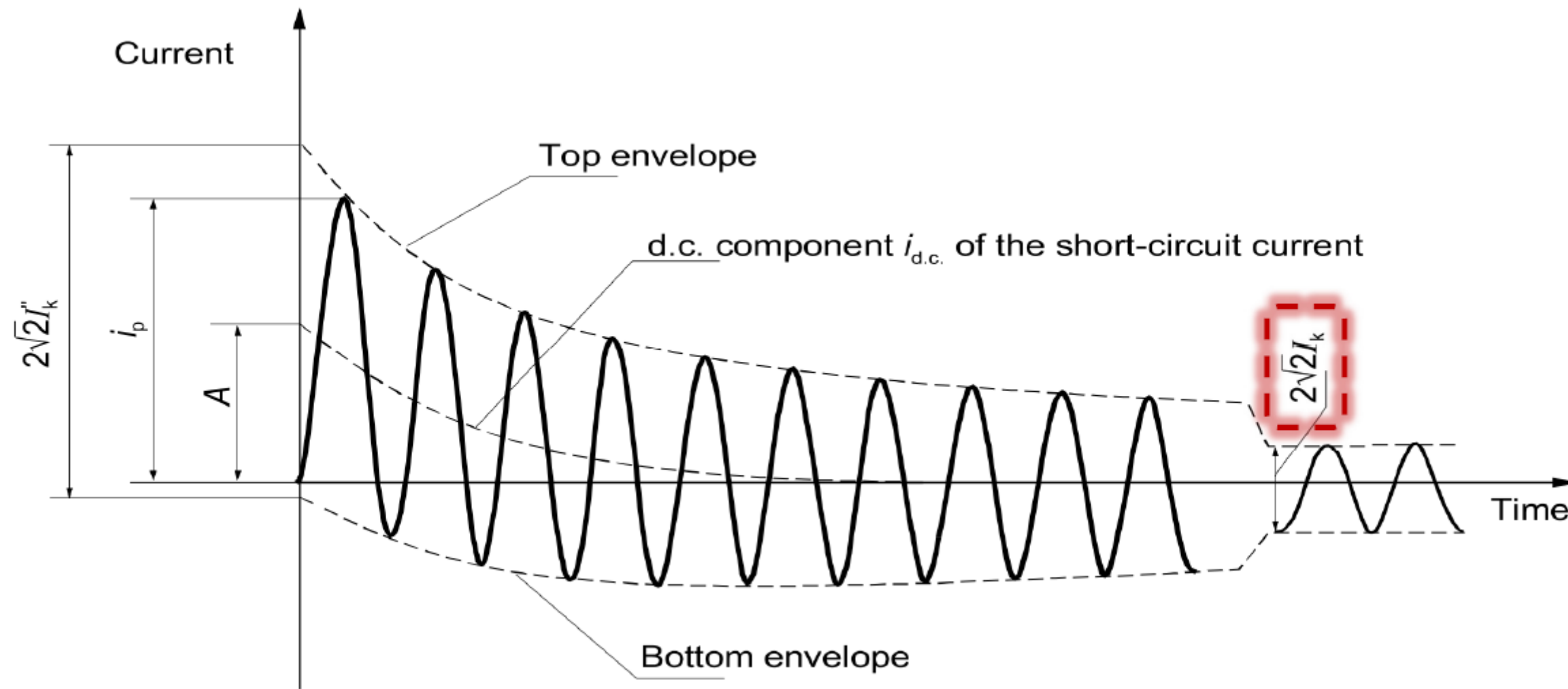


- ◆ Peak short-circuit current i_p
 - Maximum possible instantaneous value of the prospective short-circuit current

Peak short-circuit current I_P

- ♦ Affects on the mechanical stress of electrical equipment
- ♦ Can cause busbar breaks etc. mechanical damages
- ♦ i_P value is needed to ensure the protection equipment making capacity

Characteristics of short-circuit currents



- ◆ Steady-state short-circuit current I_k
 - R.m.s. value of the short-circuit current which remains after the decay of the transient phenomena
 - Depends on e.g. synchronous machines excitation

Arc flash

- ◆ Air is ionized and makes a non-zero path for the short-circuit current
- ◆ Arc flash sc-current depends on the arc resistance and can be e.g. only 60 % of the bolted sc-current
- ◆ Arc flash
 - Bright light
 - Loud noise (e.g. 130 dB)
 - Pressure shock
 - Heat (e.g. 20 000 °C)
 - Vaporized metal

Arc flash



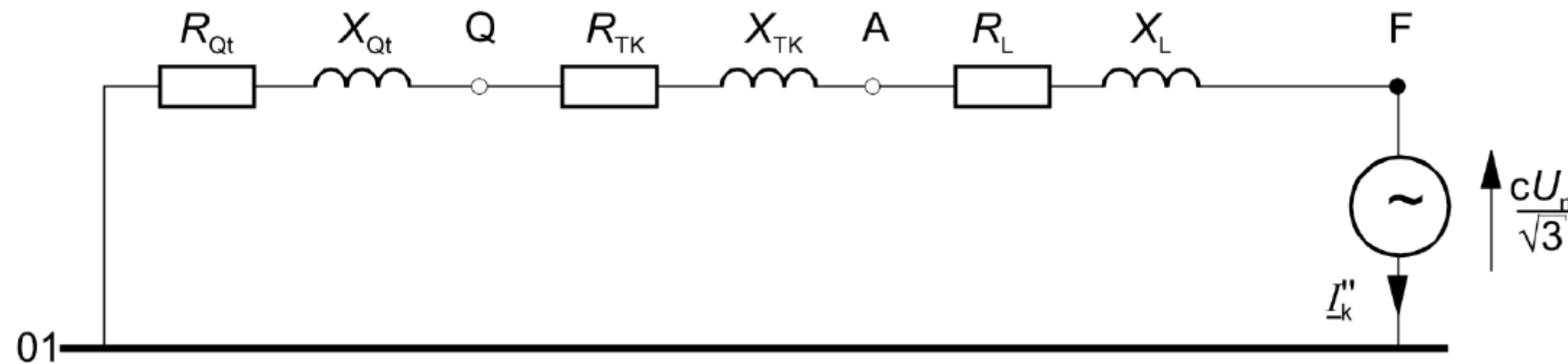
(Puttonen 2016)

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Equivalent circuit of the network

- ♦ IEC 60909-0 establishes a general procedure leading to results, which are generally of acceptable accuracy
- ♦ The method is based on the equivalent voltage source at the short-circuit location F (faulted node)
- ♦ Network feeders, synchronous and asynchronous machines, busbars, cables, overhead lines and transformers are replaced by their internal impedances

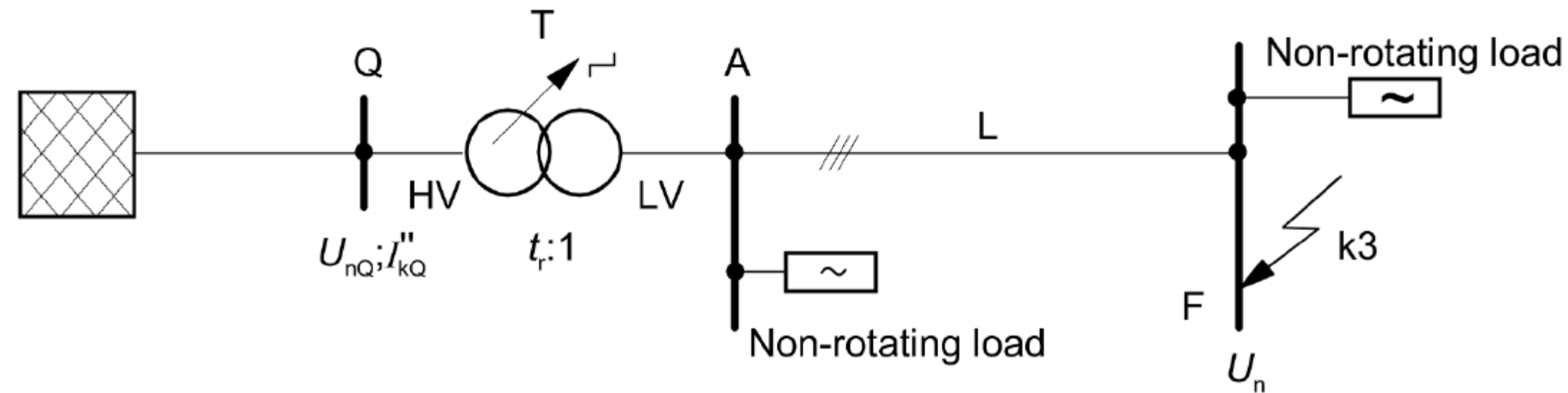


Equivalent circuit of the network

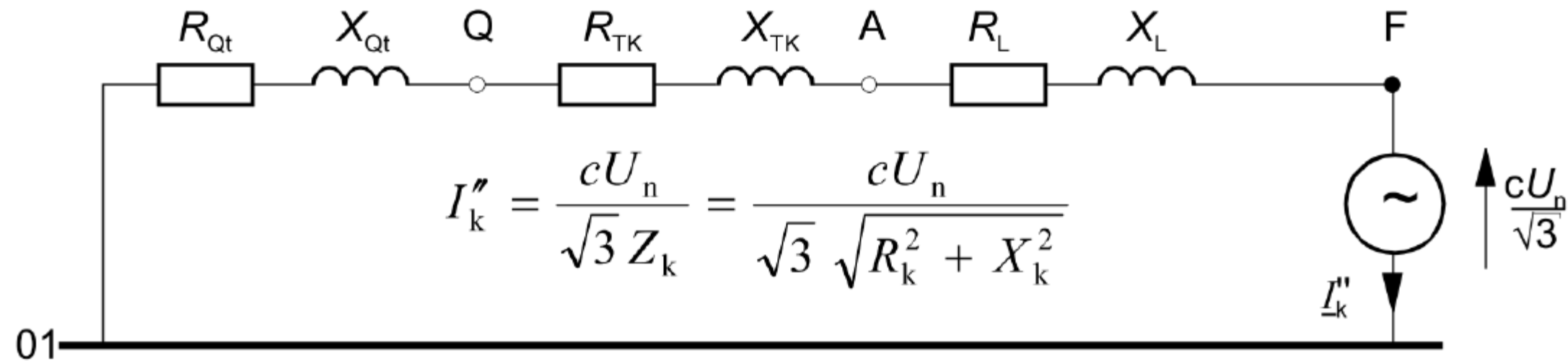
- ◆ Each network component is first represented by a equivalent circuit, which consists of
 - Voltage source, if component feeds short-circuit current to faulted node
 - Resistance R and reactance X in the "line conductor" and zero impedance in "returning conductor"
- ◆ All voltage sources are then combined to one single equivalent voltage source in the faulted node F
- ◆ In all cases it is possible to determine the shortcircuit current at the short-circuit location F with the help of an equivalent voltage source

Equivalent circuit of the network

The electrical network...



...is represented by a equivalent circuit



Equivalent circuit of the network

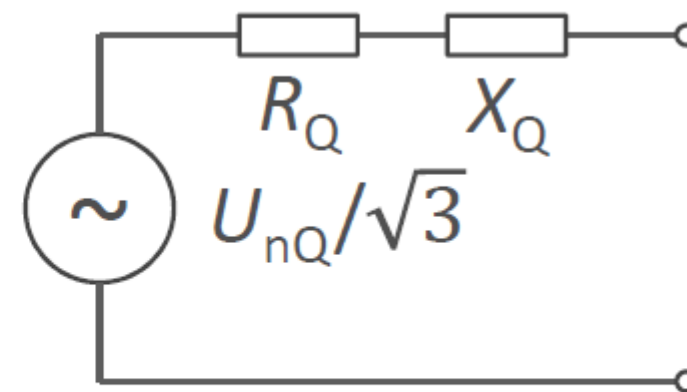
- ♦ The calculation of the short-circuit impedance is in general based on the rated data of the electrical equipment and the topological arrangement of the system
- ♦ The calculation of the short-circuit impedance has the advantage of being possible both for existing systems and for systems at the planning stage
- ♦ According to the calculation results, it is possible to make necessary improvements in the electrical network

Network feeders

- ◆ Network feeder feeds current to the faulted node
- ◆ The 1-phase equivalent circuit consists of voltage source with series resistance R_Q and reactance X_Q , that corresponds to sum of network component resistances and reactances from power plant to feeder connection point



System diagram



1-phase equivalent circuit

Network feeders

$$Z_Q = \frac{c \cdot U_{nQ}}{\sqrt{3} \cdot I''_{kQ}} = \frac{c \cdot U_{nQ}^2}{S''_{kQ}}$$

Z_Q = equivalent impedance of the network feeder (Ω)

$X_Q = 0.995 \cdot Z_Q$ = equivalent reactance of the network (Ω)

$R_Q = 0.1 \cdot X_Q$ = equivalent resistance of the network (Ω)

c = voltage factor for the voltage U_{nQ} (see next page)

U_{nQ} = nom. system voltage at the feeder connection point (V)

I''_{kQ} = symmetrical sc-current at the feeder connection point (A)

S''_{kQ} = short-circuit power at the feeder connection point (VA)

Voltage factor c for the Voltage U_n

Nominal voltage U_n	Voltage factor c for the calculation of	
	maximum short-circuit currents $c_{\max}^{1)}$	minimum short-circuit currents $c_{\min}$
Low voltage 100 V to 1 000 V (IEC 60038, table I)	1,05 ³⁾ 1,10 ⁴⁾	0,95
Medium voltage >1 kV to 35 kV (IEC 60038, table III)	1,10	1,00
High voltage²⁾ >35 kV (IEC 60038, table IV)		
¹⁾ $c_{\max}U_n$ should not exceed the highest voltage U_m for equipment of power systems. ²⁾ If no nominal voltage is defined $c_{\max}U_n = U_m$ or $c_{\min}U_n = 0,90 \times U_m$ should be applied. ³⁾ For low-voltage systems with a tolerance of +6 %, for example systems renamed from 380 V to 400 V. ⁴⁾ For low-voltage systems with a tolerance of +10 %.		

Network feeders

Example: At the feeder connection point nominal system voltage is $U_{nQ} = 0.4$ kV, low-voltage system tolerance is 6 % and maximum symmetrical short-circuit current is $I''_{kQ} = 47.6$ kA. Calculate equivalent Z_Q , X_Q and R_Q of the network feeder.

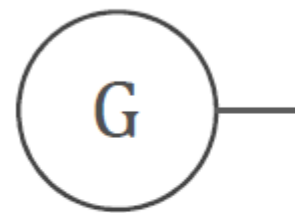
$$Z_Q = \frac{c \cdot U_{nQ}}{\sqrt{3} \cdot I''_{kQ}} = \frac{1.05 \cdot 400 \text{ V}}{\sqrt{3} \cdot 47600 \text{ A}} = 0.0051 \text{ } \Omega$$

$$X_Q = 0.995 \cdot Z_Q = 0.995 \cdot 0.0051 \text{ } \Omega = 0.00507 \text{ } \Omega$$

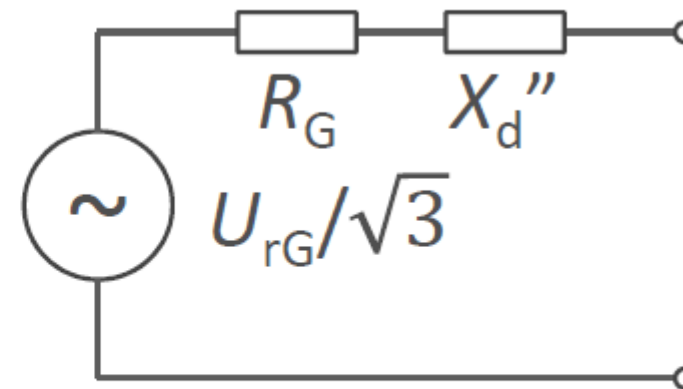
$$R_Q = 0.1 \cdot X_Q = 0.1 \cdot 0.00507 \text{ } \Omega = 0.00051 \text{ } \Omega$$

Synchronous machines

- ♦ Synchronous machine (generator) feeds current to the faulted node
- ♦ The 1-phase equivalent circuit consists of voltage source with series resistance R_G and reactance X_d'' that corresponds to winding resistance and reactance of the machine



System diagram



1-phase equivalent circuit

Synchronous machines

$$X_d'' = K_G \cdot \frac{x_d''}{100 \%} \cdot \frac{U_{rG}^2}{S_{rG}}, K_G = \frac{U_n}{U_{rG}} \cdot \frac{c_{\max}}{1 + \frac{x_d''}{100 \%} \sin \varphi_{rG}}$$

K_G is used, when calculating initial symmetrical short-circuit currents in systems fed directly from synchronous machines without unit transformers, e.g. in industrial or in LV networks

X_d'' = subtransient reactance of the synchronous machine (Ω)

x_d'' = relative subtransient reactance of the sm (%)

U_{rG} = rated voltage of the synchronous machine (V)

S_{rG} = rated apparent power of the synchronous machine (VA)

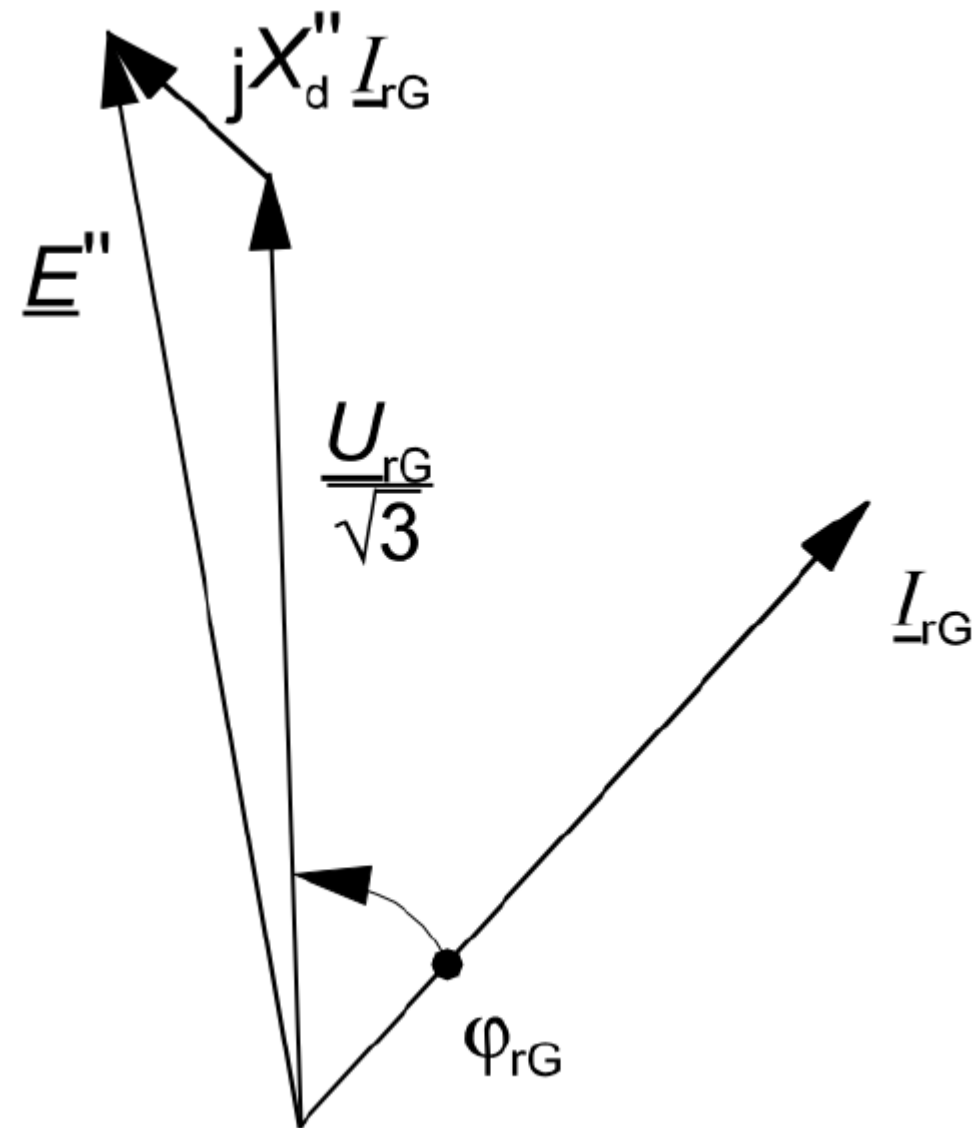
U_n = nominal voltage of the system (V)

$c_{\max}$ = voltage factor for the voltage U_n

φ_{rG} = phase angle between $\underline{I}_{rG}$ and $\underline{U}_{rG}/\sqrt{3}$ ($^\circ$)

$R_G \approx 0$ on medium and high power machines

Synchronous machines



Synchronous machines

MAIN STATOR RESISTANCE	0.0016 Ohms PER PHASE AT 22°C STAR CONNECTED								
MAIN ROTOR RESISTANCE	1.67 Ohms at 22°C								
EXCITER STATOR RESISTANCE	17.5 Ohms at 22°C								
EXCITER ROTOR RESISTANCE	0.063 Ohms PER PHASE AT 22°C								
R.F.I. SUPPRESSION	BS EN 61000-6-2 & BS EN 61000-6-4,VDE 0875G, VDE 0875N. refer to factory for others								
WAVEFORM DISTORTION	NO LOAD < 1.5% NON-DISTORTING BALANCED LINEAR LOAD < 5.0%								
MAXIMUM OVERSPEED	2250 Rev/Min								
BEARING DRIVE END	BALL. 6228 C3								
BEARING NON-DRIVE END	BALL. 6319 C3								
		50 Hz				60 Hz			
TELEPHONE INTERFERENCE	THF<2%				TIF<50				
COOLING AIR	2.69 m³/sec 50 m³/min				3.45 m³/sec 7300 cfm				
VOLTAGE STAR	380/220	400/231	415/240	440/254	416/240	440/254	460/266	480/277	
kVA BASE RATING FOR REACTANCE VALUES	1140	1175		1150	1275	1395	1405	1415	
Xd DIR. AXIS SYNCHRONOUS	3.26	3.04	2.82	2.45	3.93	3.85	3.55	3.28	
X'd DIR. AXIS TRANSIENT	0.20	0.19	0.17	0.15	0.24	0.23	0.22	0.20	
X''d DIR. AXIS SUBTRANSIENT	0.15	0.14		0.11	0.18	0.17	0.16	0.15	

Synchronous machines

Example: Rated voltage of the generator connected to $U_n = 0.4$ kV network without unit transformer is $U_{rG} = 0.4$ kV, rated apparent power of the generator is $S_{rG} = 1175$ kVA and relative subtransient reactance of the generator is $x_d'' = 14\%$. Phase angle between I_{rG} and $\underline{U}_{rG}/\sqrt{3}$ is $\varphi_{rG} = 30^\circ$. Calculate subtransient reactance X_d'' (ohmic value).

$$K_G = \frac{U_n}{U_{rG}} \cdot \frac{c_{\max}}{1 + \frac{x_d''}{100\%} \sin \varphi_{rG}} = \frac{400 \text{ V}}{400 \text{ V}} \cdot \frac{1.05}{1 + \frac{14\%}{100\%} \sin 30^\circ} = 0.98$$

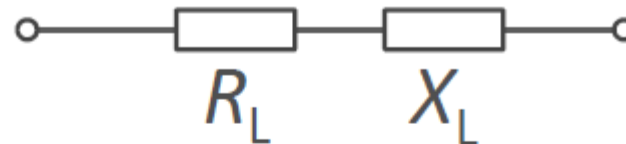
$$X_d'' = K_G \cdot \frac{x_d''}{100\%} \cdot \frac{U_{rG}^2}{S_{rG}} = 0.98 \cdot \frac{14\%}{100\%} \cdot \frac{(400 \text{ V})^2}{1.175 \cdot 10^6 \text{ VA}} = 0.019 \Omega$$

Busbars, cables or overhead lines

- ♦ Busbar, cable or overhead line is passive component in the electrical network; it attenuates short-circuit current
- ♦ The 1-phase equivalent circuit consists only of resistance R_L and reactance X_L , that corresponds to conductor resistance and reactance



System diagram



1-phase equivalent circuit

Busbars, cables or overhead lines

- ♦ R_L depends on conductor length, conductor material, cross-sectional area, length, number of parallel connected conductors and temperature of the conductor
- ♦ X_L depends on conductor length, number of parallel connected conductors and the centre-distances of the conductors
- ♦ The conductor material and temperature of the conductor has no effect on the reactance X_L

Busbars, cables or overhead lines

$$R_L = \frac{R'_L \cdot l}{n}$$

$$X_L = \frac{X'_L \cdot l}{n} = \frac{2\pi \cdot f \cdot L_b \cdot l}{n}$$

R_L = resistance of the conductor (Ω)

X_L = reactance of the conductor (Ω)

R'_L = resistance per unit length (Ω/km)

X'_L = reactance per unit length (Ω/km)

l = length of the conductor (km)

n = number of parallel connected conductors (pcs)

f = frequency (Hz), typically 50 Hz

L_b = inductance per unit length (H/km)

Busbars, cables or overhead lines

When calculating minimum short-circuit currents, it is necessary to introduce resistances R_L of lines at a higher temperature:

$$R_L = [1 + \alpha(\theta_e - 20^\circ\text{C})] \cdot R_{L20}$$

R_{L20} = resistance at a temperature of 20 °C

θ_e = conductor temperature in degrees Celsius at the end of the short-circuit duration; in low voltage 80 °C

α = factor equal to 0.004/K, valid with sufficient accuracy for most practical purposes for copper, aluminium and aluminium alloy

Busbars, cables or overhead lines

PRODUCT NAME			AMCMK 3x16/10 1 kV	AMCMK 3x25/16 1 kV	AMCMK 3x35/16 AN 1 kV	AMCMK 3x50/16 AN 1 kV	AMCMK 3x70/21 AN 1 kV	AMCMK 3x95/29 AN 1 kV	AMCMK 3x120/41 AN 1 kV
STK-code			0622157	0622158	0622159	0622160	0622161	0622162	0622163
ELECTRICAL DATA (2)									
Maximum DC resistance of phase conductor.	conductor 20°C	Ω/km	1,91	1,20	← 0,868	R'_L 0,641	0,443	0,320	0,253
AC resistance of phase and neutral conductor (1)	conductor 70°C	Ω/km	2,3	1,4	1,0	0,77	0,53	0,39	0,31
Maximum DC resistance of PE conductor	conductor 20°C	Ω/km	1,83	1,15	1,15	1,15	0,868	0,641	0,443
Inductance (1)		mH/km	0,28	0,28	← 0,26	L_b 0,25	0,24	0,24	0,23

(Puttonen 2016)

Busbars, cables or overhead lines

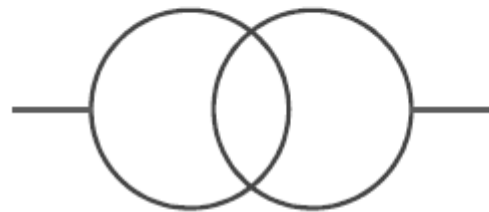
Example: Resistance per unit length of the conductor at a temperature of 20 °C is $R'_L = 0.320 \Omega/\text{km}$, inductance per unit length is $L_b = 0.24 \text{ mH/km}$ and length of the conductor is $l = 0.250 \text{ km}$. Calculate total resistance and reactance of two parallel connected conductors, when frequency is $f = 50 \text{ Hz}$.

$$R_L = \frac{R'_L \cdot l}{n} = \frac{0.320 \frac{\Omega}{\text{km}} \cdot 0.250 \text{ km}}{2} = 0.040 \Omega$$

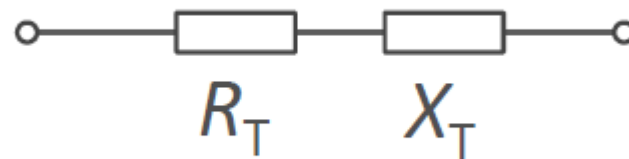
$$X_L = \frac{X'_L \cdot l}{n} = \frac{2\pi \cdot f \cdot L_b \cdot l}{n} = \frac{2\pi \cdot 50 \frac{1}{\text{s}} \cdot 0.00024 \frac{\text{H}}{\text{km}} \cdot 0.250 \text{ km}}{2} = 0.0094 \Omega$$

Transformers

- ♦ Transformer is passive component in the electrical network; it attenuates short-circuit current
- ♦ The 1-phase equivalent circuit consists only of resistance R_T and reactance X_T , that corresponds to short-circuit resistance and reactance of transformer



System diagram



1-phase equivalent circuit

Transformers

$$Z_T = \frac{u_{kr}}{100 \%} \cdot \frac{U_{rT}^2}{S_{rT}}$$

$$R_T = \frac{u_{Rr}}{100 \%} \cdot \frac{U_{rT}^2}{S_{rT}} = \frac{P_{krT}}{3I_{rT}^2} = P_{krT} \cdot \frac{U_{rT}^2}{S_{rT}^2}$$

$$X_T = \sqrt{Z_T^2 - R_T^2}$$

Z_T , R_T and X_T = short-circuit impedance, resistance and reactance (Ω)

u_{kr} = short-circuit voltage at rated current (%)

U_{rT} = rated voltage of the transformer on the HV or LV side (V)

S_{rT} = rated apparent power of the transformer (VA)

u_{Rr} = relative resistive component of the short-circuit voltage (%)

P_{krT} = total active power loss in the windings at rated current (W)

I_{rT} = rated current of the transformer on the HV or LV side (A)

Transformers

110kV Three Phases Double Windings Power Transformer with On Load Tap Changer

Rated capacity (kVA)	Voltage ratio (kV)		Vector group	No-load loss (kW)	Load loss (kW)	No-load current (%)	Short- circuit impedance (%)	Weight (MT)		Dimension (mm)		
	HV	LV						Oil	Total	L	B	H
S_{rT}	U_{rT}		Ynd11	7	P_{krT}	0.3	u_{kr}	6.6	22.5	5200	3700	4300
10000	110±8× 1.25%			9	50			7.1	25.2	5420	3950	4420
12500				10				7.6	28	5930	4280	4560
16000				12				8.3	31.8	5990	4310	4670
20000				14				9.7	36.4	6320	4340	4810
25000				18	88	10.8	40.6	6400	4380	4820		
31500				20	104	11.3	45.8	6530	4420	4990		
40000				23	125	12.8	51	6580	4510	5080		
50000				25	146	14.8	60.7	6700	4810	5280		
63000				30	180	16.2	69	6860	5080	5320		
				38	220	17.5	77.6	6960	5100	5610		

Transformers

Example: Total active power loss in the windings at rated current is $P_{krT} = 14.6$ kW, rated apparent power is $S_{rT} = 2000$ kVA, short-circuit voltage at rated current is $u_{krT} = 6.0$ % and rated transformation ratio is $U_{1r}/U_{2r} = 10/0.4$ kV. Calculate Z_T , R_T and X_T on 0.4 kV voltage level.

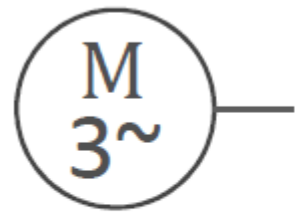
$$Z_T = \frac{u_{kr}}{100 \%} \cdot \frac{U_{rT}^2}{S_{rT}} = \frac{6.0 \%}{100 \%} \cdot \frac{(400 \text{ V})^2}{2 \cdot 10^6 \text{ VA}} = 0.0048 \Omega$$

$$R_T = P_{krT} \cdot \frac{U_{rT}^2}{S_{rT}^2} = 14600 \text{ W} \cdot \frac{(400 \text{ V})^2}{(2 \cdot 10^6 \text{ VA})^2} = 0.000584 \Omega$$

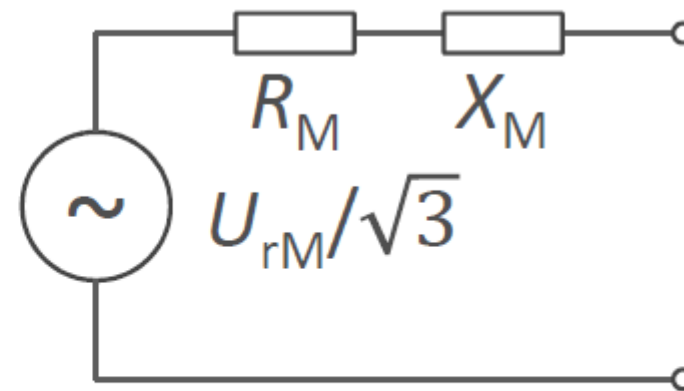
$$X_T = \sqrt{Z_T^2 - R_T^2} = \sqrt{0.0048^2 - 0.000584^2} \Omega = 0.004764 \Omega$$

Asynchronous motors

- ♦ Asynchronous motor feeds decaying short-circuit current to the faulted node at the beginning of the sc
- ♦ The 1-phase equivalent circuit consists of voltage source with series resistance R_M and reactance X_M , that corresponds to winding resistance and reactance of the motor



System diagram



1-phase equivalent circuit

Asynchronous motors

- ♦ Low-voltage motors are usually connected to the busbar by cables with different lengths and cross-sections
- ♦ For simplification of the calculation, groups of motors including their connection cables may be combined to a single equivalent motor (represented without cable)
- ♦ When using single equivalent motor, the ratio of the locked-rotor current to the rated current is assumed to be $I_{LR}/I_{rM} = 5$

Asynchronous motors

$$Z_M = \frac{1}{I_{LR}/I_{rM}} \cdot \frac{U_{rM}}{\sqrt{3} \cdot I_{rM}} = \frac{1}{I_{LR}/I_{rM}} \cdot \frac{U_{rM}^2}{S_{rM}}$$

$I_{LR}/I_{rM} = 5$, ratio of the locked-rotor current to the rated current

U_{rM} = rated voltage of the motor (V)

I_{rM} = sum of the rated currents of the motors in a group (A)

S_{rM} = sum of the apparent powers of the motors in a group (VA)

$$X_M = 0.922 \cdot Z_M, R_M = 0.42 \cdot X_M$$

Asynchronous motors

Example: Group of low-voltage motors are connected to $U_n = 0.4$ kV network. Sum of the apparent powers of the motors in a group is $S_{rM} = 0.585$ MVA. Calculate Z_M , R_M ja X_M .

$$Z_M = \frac{1}{\frac{I_{LR}}{I_{rM}}} \cdot \frac{U_{rM}^2}{S_{rM}} = \frac{1}{5} \cdot \frac{(400 \text{ V})^2}{585000 \text{ VA}} = 0.0547 \text{ } \Omega$$

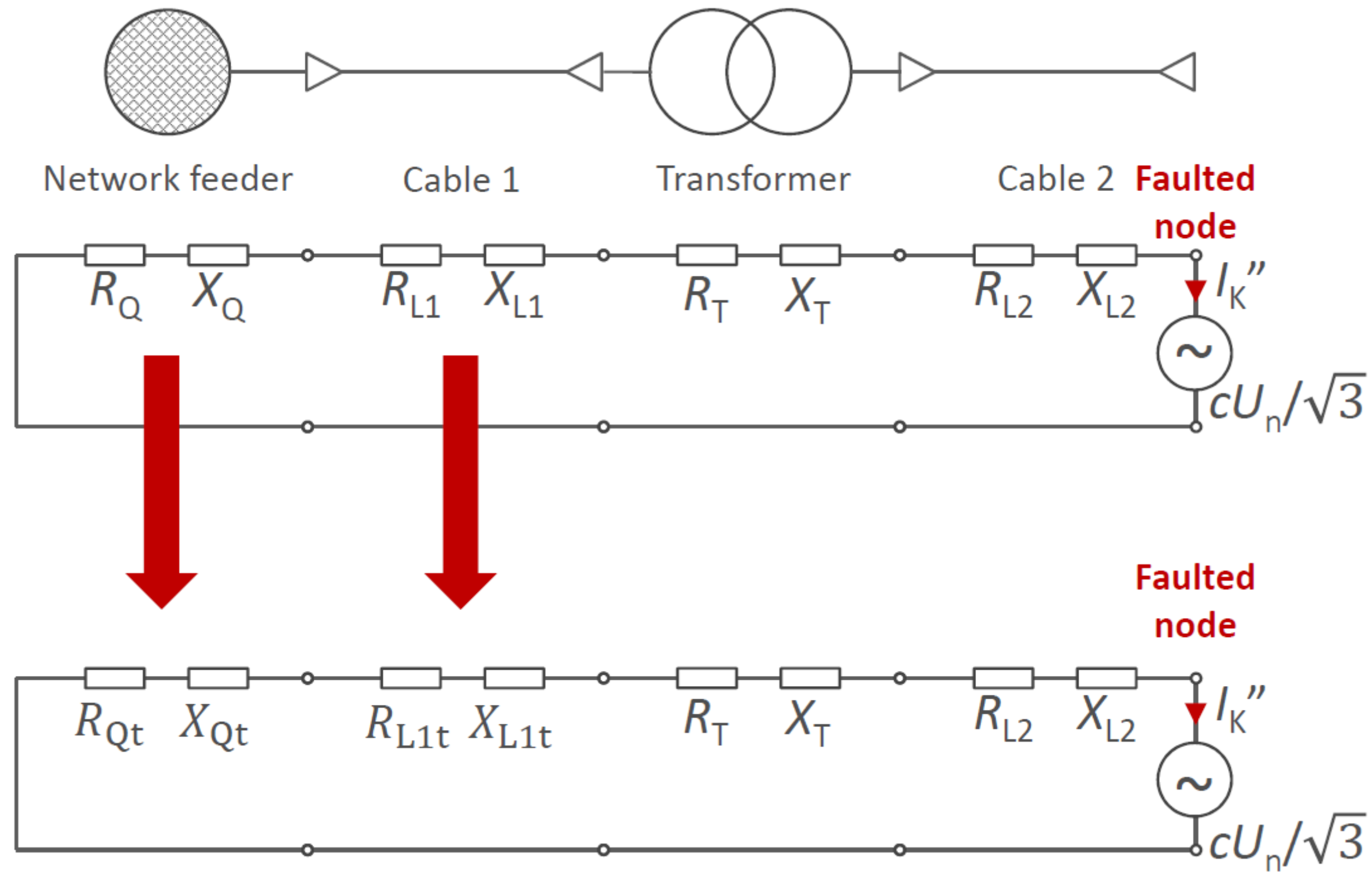
$$X_M = 0.922 \cdot Z_M = 0.922 \cdot 0.0547 \text{ } \Omega = 0.0504 \text{ } \Omega$$

$$R_M = 0.42 \cdot X_M = 0.42 \cdot 0.0504 \text{ } \Omega = 0.0212 \text{ } \Omega$$

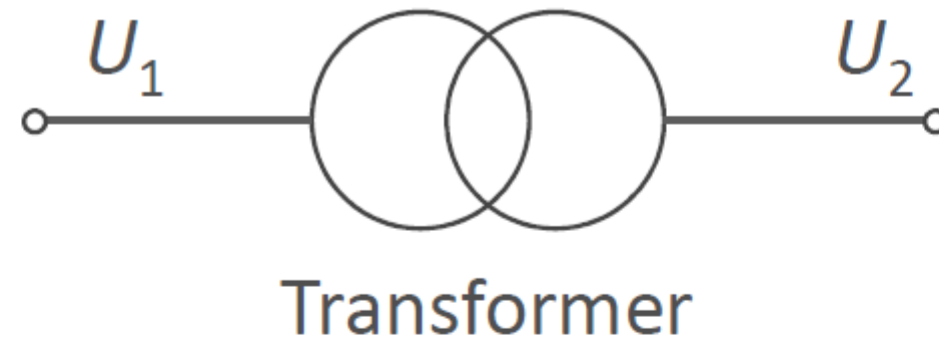
Impedance transfer

- ◆ When calculating short-circuit currents in systems with different voltage levels, it is necessary to transfer impedance values from one voltage level to another, usually to that voltage level at which the short-circuit current is to be calculated
- ◆ Transferred quantity is marked with "t", e.g. R_{Qt}
- ◆ The impedances of the equipment in superimposed or subordinated networks are to be divided or multiplied by the square of the rated transformation ratio $t_r = U_{1r}/U_{2r}$

Impedance transfer



Impedance transfer



Transferring impedance from voltage level U_2 to U_1 :

$$Z_{2t} = t_r^2 \cdot Z_2 = \left(\frac{U_{1r}}{U_{2r}} \right)^2 \cdot Z_2$$

Transferring impedance from voltage level U_1 to U_2 :

$$Z_{1t} = \frac{1}{t_r^2} \cdot Z_1 = \left(\frac{U_{2r}}{U_{1r}} \right)^2 \cdot Z_1$$

Impedance transfer

Example: Network feeder equivalent impedance on the transformer high voltage (10 kV) level is $Z_{Q10kV} = 0.2 \Omega$. Calculate network feeder equivalent impedance transferred to the low voltage (0.4 kV) voltage level.

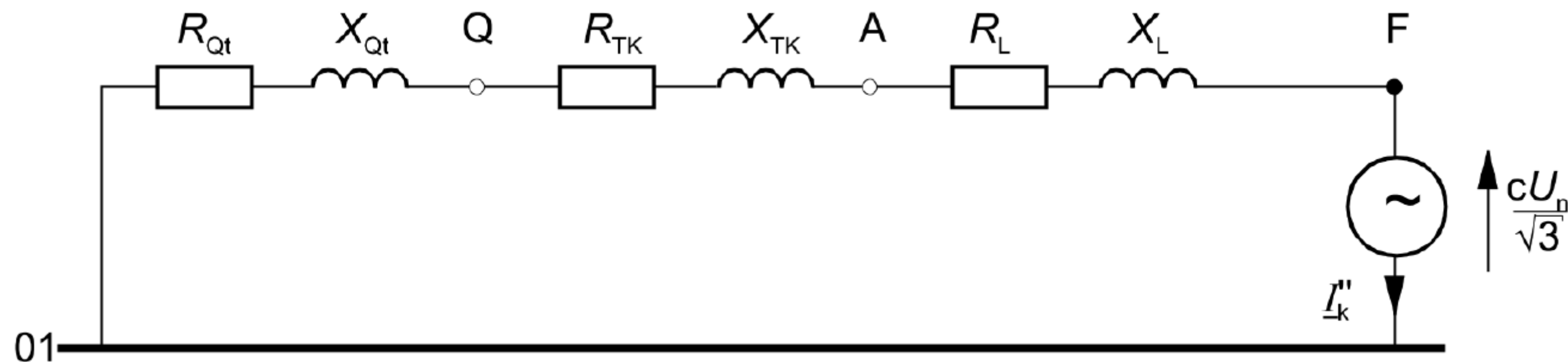
$$Z_{Q10kVt} = Z_{Q10kV} \cdot \left(\frac{U_{2r}}{U_{1r}} \right)^2 = 0.2 \Omega \cdot \left(\frac{0.4 \text{ kV}}{10 \text{ kV}} \right)^2 = 0.00032 \Omega$$

Content

- ◆ Introduction
 - ◆ Definition of short circuit, different types of short circuits, characteristics of the current
- ◆ Short-circuit impedances of electrical equipment
 - ◆ Equivalent circuits, calculation of resistance and reactance of each component, examples, exercises
- ◆ **Calculation of short circuit currents**
 - ◆ **Calculation assumptions, formulas, examples, exercises**

Method of calculation

- ♦ The method is based on the equivalent voltage source at the short-circuit location F (faulted node)
- ♦ The equivalent voltage source is the only active voltage of the system
- ♦ All network feeders, synchronous and asynchronous machines are replaced by their internal impedances



Calculation of short-circuit currents

- ♦ In general, three short-circuit currents, which differ in their magnitude, are to be calculated:
 - The maximum short-circuit current which determines the capacity or rating of electrical equipment
 - The steady-state short-circuit current which is needed to calculate the thermal effect of the decaying short-circuit current
 - The minimum short-circuit current which can be a basis for example, for the selection of fuses and for the setting of protective devices

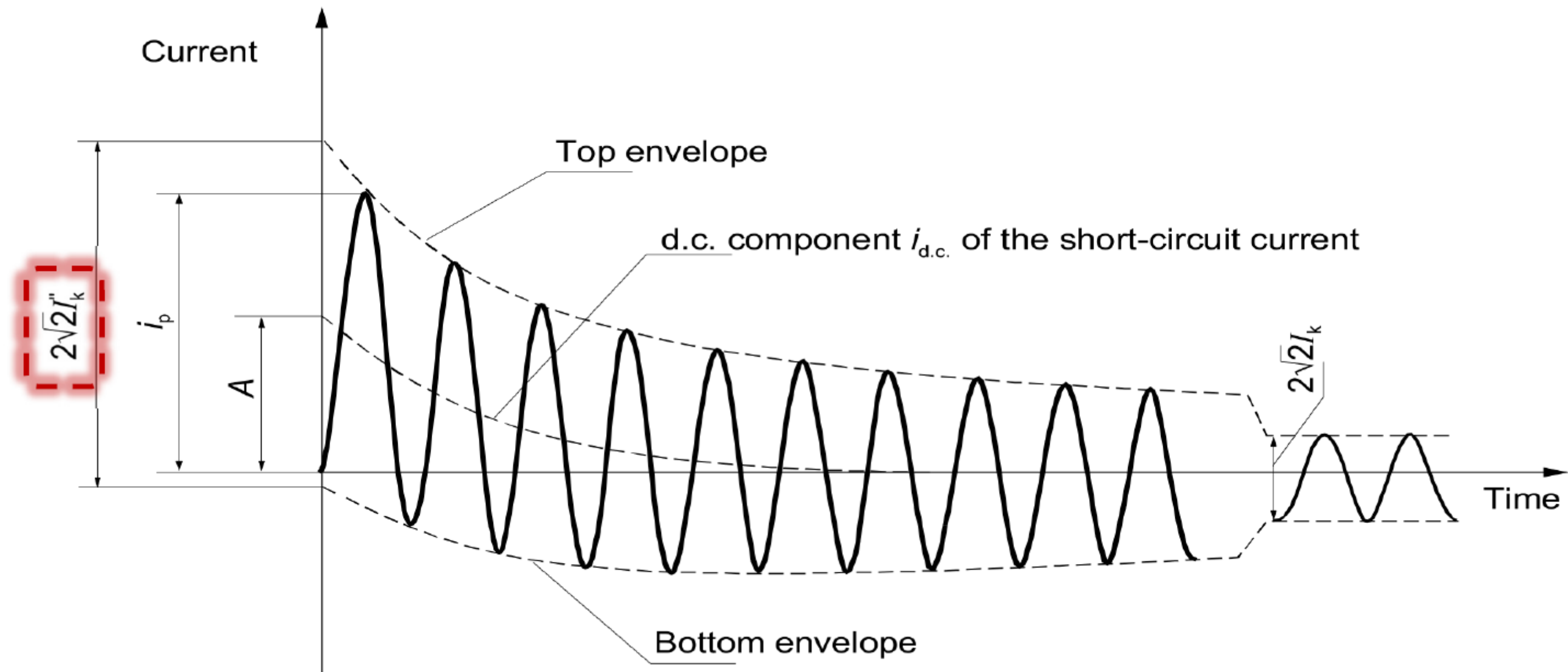
Calculation assumptions

- ♦ The calculation of maximum, steady state and minimum short-circuit currents is based on the following simplifications
 - There is no change in the type of short circuit during the time of short circuit e.g. a three-phase short circuit remains three-phase
 - There is no change in the network structure during the time of short circuit
 - Arc resistances are not taken into account

Short-circuit currents

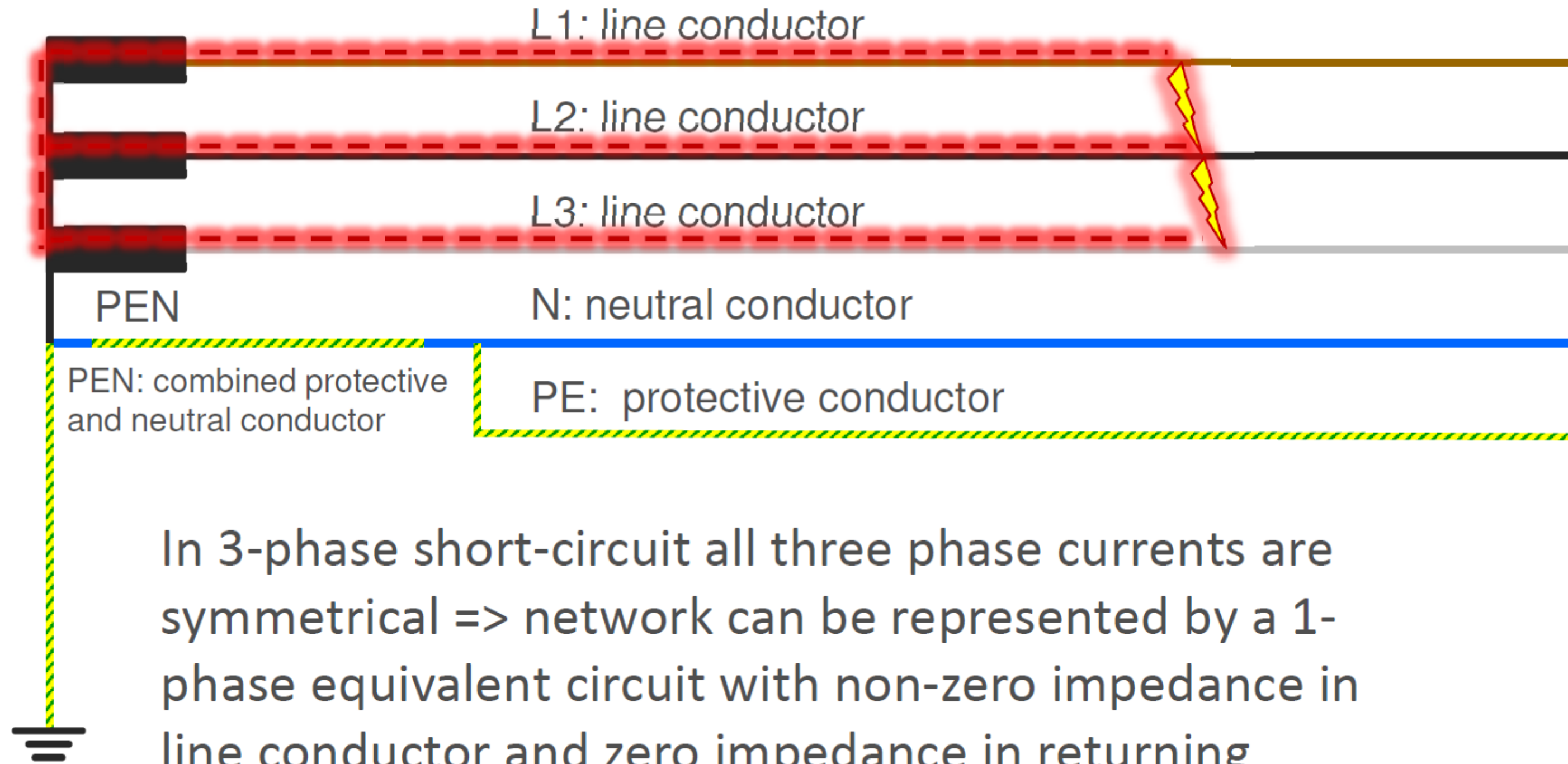
- ◆ It is necessary to know the magnitudes of the short-circuit currents
 - To select and determine the characteristics of equipment (thermal and mechanical withstand capacity)
 - To select the protection equipment breaking capacity
 - To select the protection equipment tripping settings
 - To ensure discrimination and selectivity

Maximum 3-phase short-circuit current I''_k



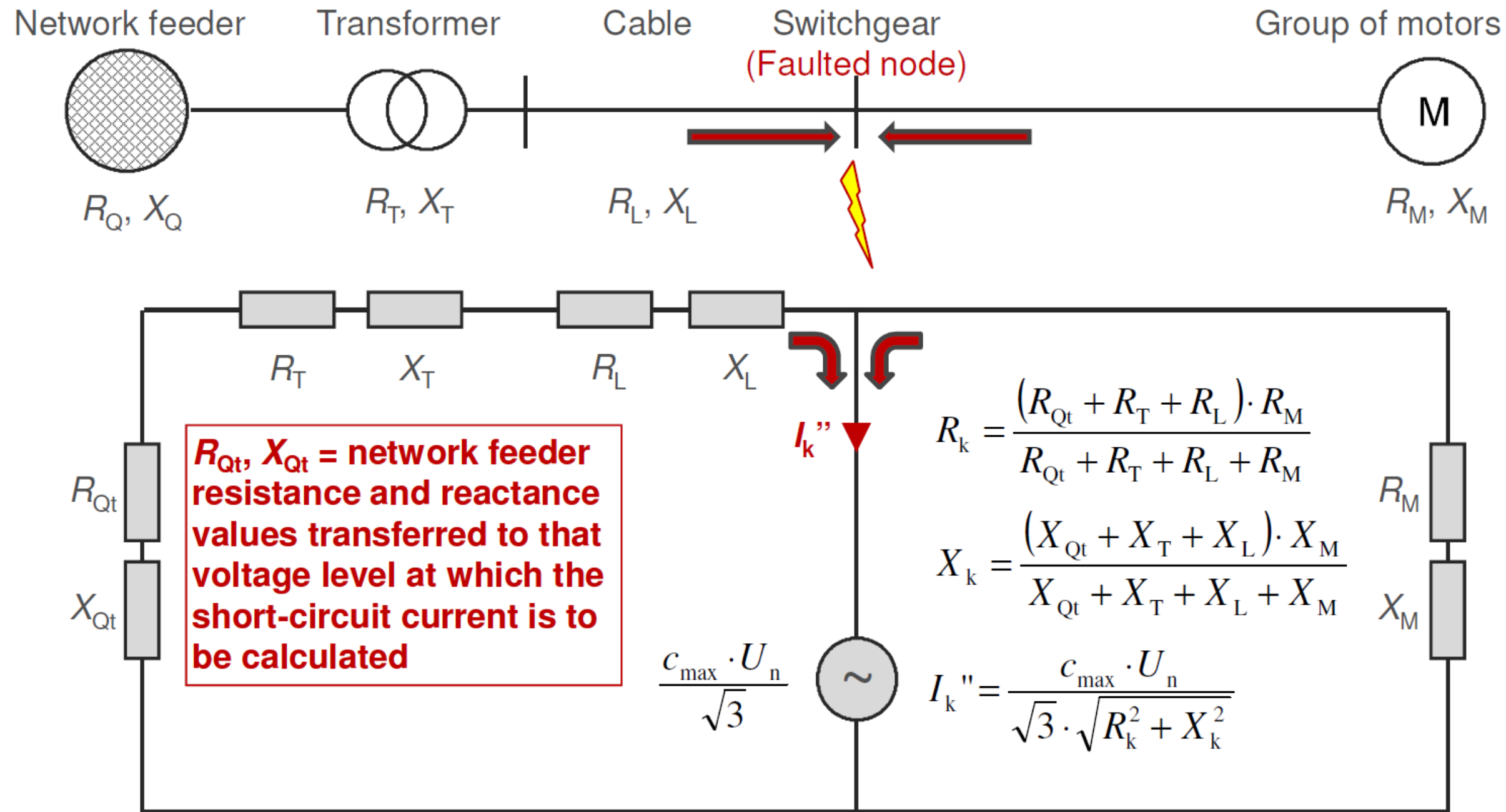
- ◆ Initial symmetrical short-circuit current I''_k
 - R.m.s. value of the a.c. symmetrical prospective shortcircuit current at the beginning of short-circuit (thermal effect on busbars)

Maximum 3-phase short-circuit current I''_k



In 3-phase short-circuit all three phase currents are symmetrical => network can be represented by a 1-phase equivalent circuit with non-zero impedance in line conductor and zero impedance in returning conductor.

Maximum 3-phase short-circuit current I''_k



Maximum 3-phase short-circuit current I''_k

- Voltage factor $c_{\max}$:
 - Safety factor, depending on voltage level
 - $0.1 \text{ kV} < U_n \leq 1 \text{ kV}$ $c_{\max} = 1.05 \dots 1.10$ (see next page)
 - $U_n > 1 \text{ kV}$ $c_{\max} = 1.10$
- When calculating maximum short-circuit current:
 - Resistance R_L of lines obtained at a temperature of 20°C
 - Fault impedance Z_F assumed to be zero
 - **Motors and generators are included** (feeding current to the faulted node)
 - System configuration and the maximum contribution from power plants/network feeders which lead to the maximum value of short-circuit current at the short-circuit location

Maximum 3-phase short-circuit current I''_k

Nominal voltage U_n	Voltage factor c for the calculation of	
	maximum short-circuit currents $c_{\max}^{1)}$	minimum short-circuit currents $c_{\min}$
Low voltage 100 V to 1 000 V (IEC 60038, table I)	1,05 ³⁾ 1,10 ⁴⁾	0,95
Medium voltage >1 kV to 35 kV (IEC 60038, table III)	1,10	1,00
High voltage²⁾ >35 kV (IEC 60038, table IV)		

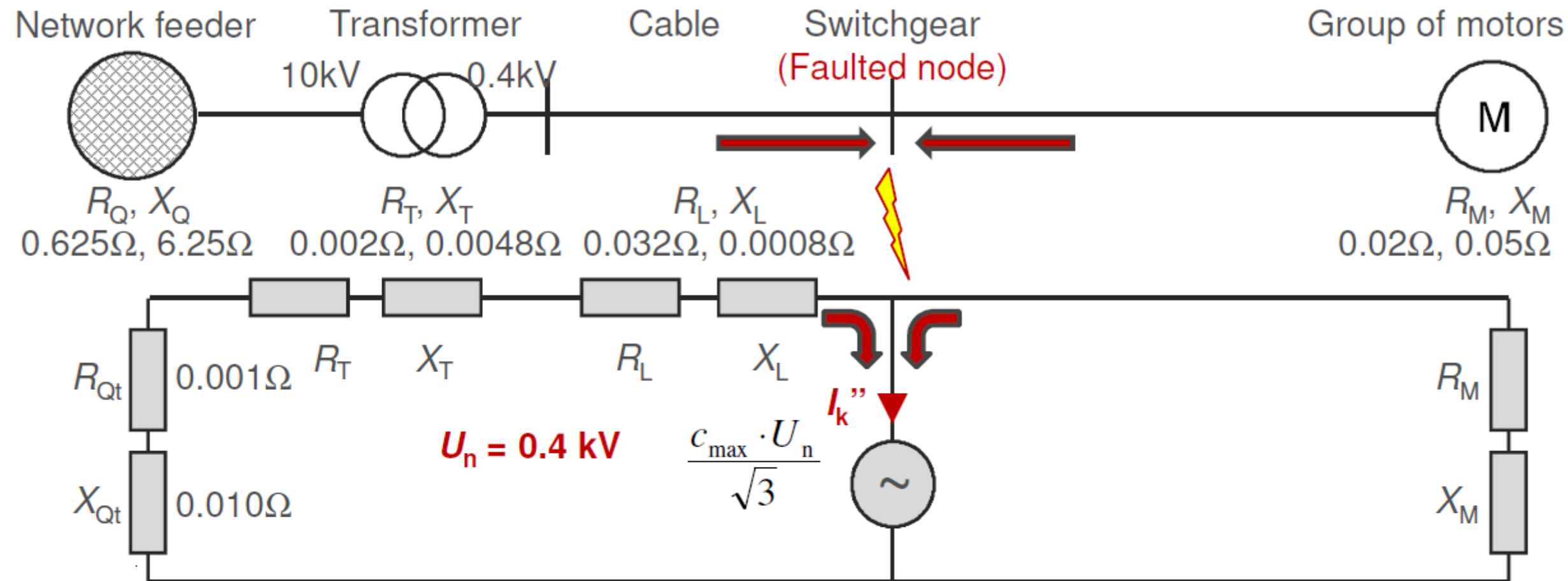
¹⁾ $c_{\max}U_n$ should not exceed the highest voltage U_m for equipment of power systems.

²⁾ If no nominal voltage is defined $c_{\max}U_n = U_m$ or $c_{\min}U_n = 0,90 \times U_m$ should be applied.

³⁾ For low-voltage systems with a tolerance of +6 %, for example systems renamed from 380 V to 400 V.

⁴⁾ For low-voltage systems with a tolerance of +10 %.

Maximum 3-phase short-circuit current I''_k

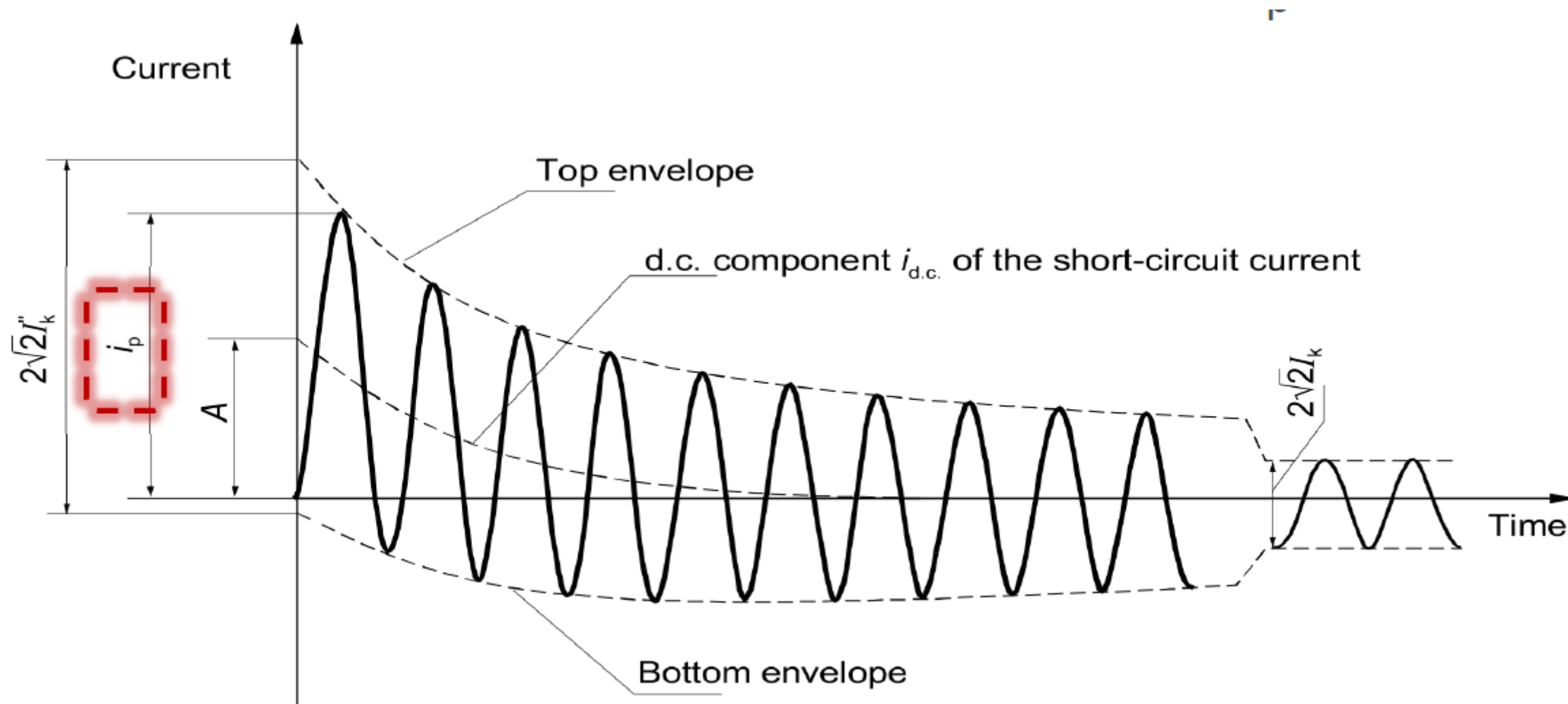


$$R_k = \frac{(R_{Qt} + R_T + R_L) \cdot R_M}{R_{Qt} + R_T + R_L + R_M} = \frac{(0.001 + 0.002 + 0.032) \cdot 0.02}{0.001 + 0.002 + 0.032 + 0.02} = 0.013 \Omega$$

$$X_k = \frac{(X_{Qt} + X_T + X_L) \cdot X_M}{X_{Qt} + X_T + X_L + X_M} = \frac{(0.010 + 0.0048 + 0.0008) \cdot 0.05}{0.010 + 0.0048 + 0.0008 + 0.05} = 0.012 \Omega$$

$$I''_k = \frac{c_{\max} \cdot U_n}{\sqrt{3} \cdot \sqrt{R_k^2 + X_k^2}} = \frac{1.05 \cdot 400 \text{ V}}{\sqrt{3} \cdot \sqrt{0.013^2 + 0.012^2}} = 13706 \text{ A} = 13.7 \text{ kA}$$

Peak short-circuit current i_p



- ◆ Peak short-circuit current i_p
 - Maximum possible instantaneous value of the prospective short-circuit current (mechanical effect on busbars)

Peak short-circuit current i_p

- In non-meshed networks i_p can be expressed by

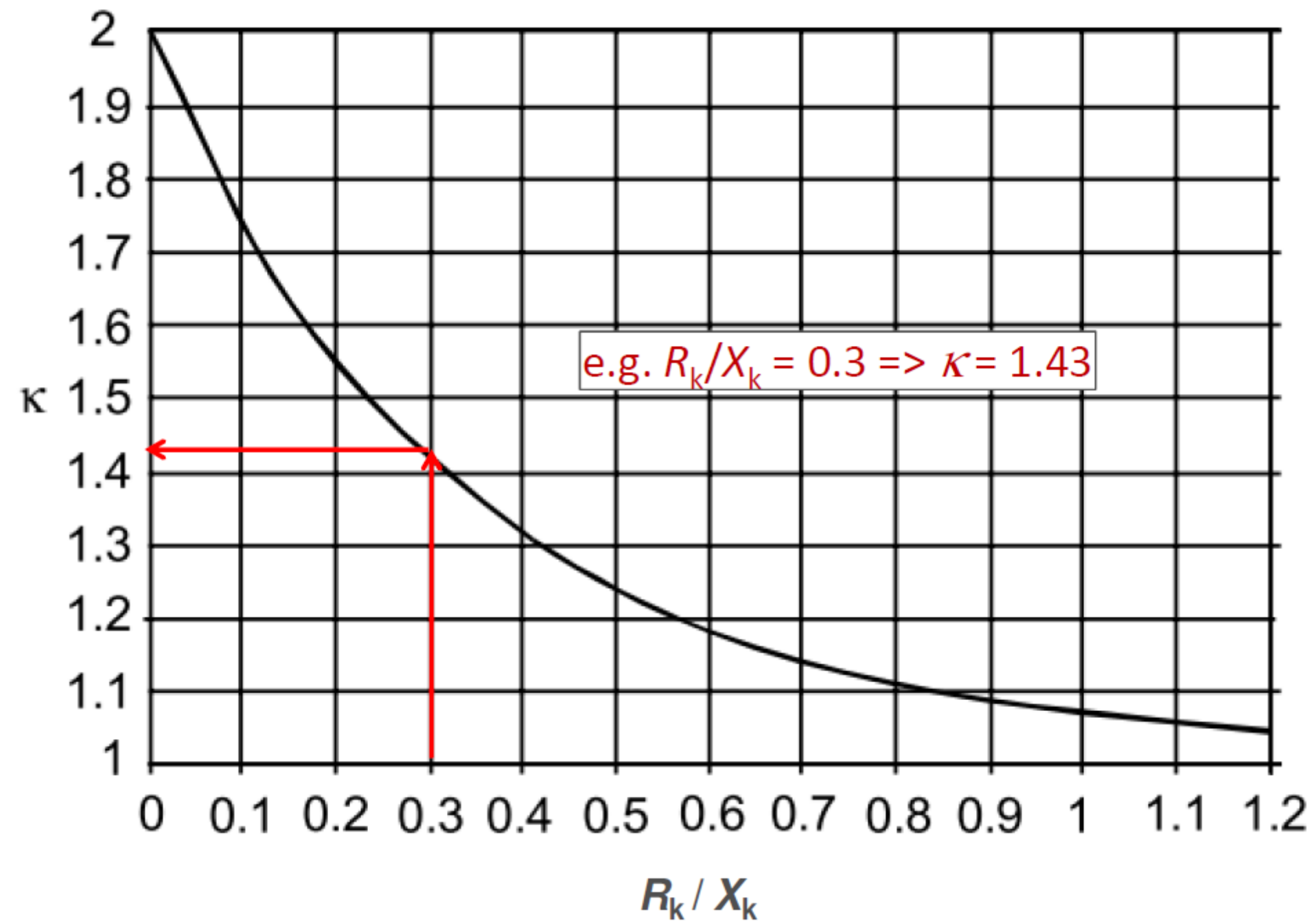
$$i_p = \kappa \cdot \sqrt{2} \cdot I_k''$$

- κ = peak factor, depends on R_k/X_k -ratio
- Peak factor κ can be obtained from figure on next page or calculated with the following formula

$$\kappa = 1.022 + 0.969 \cdot e^{\frac{-3.03 \cdot R_k}{X_k}}$$

- In low voltage networks typically is $\kappa \approx 1.4 \dots 1.5$
- In medium and high voltage networks $\kappa \approx 1.8 \dots 2.0$

Peak short-circuit current i_p



Peak short-circuit current i_p

According to previous example:

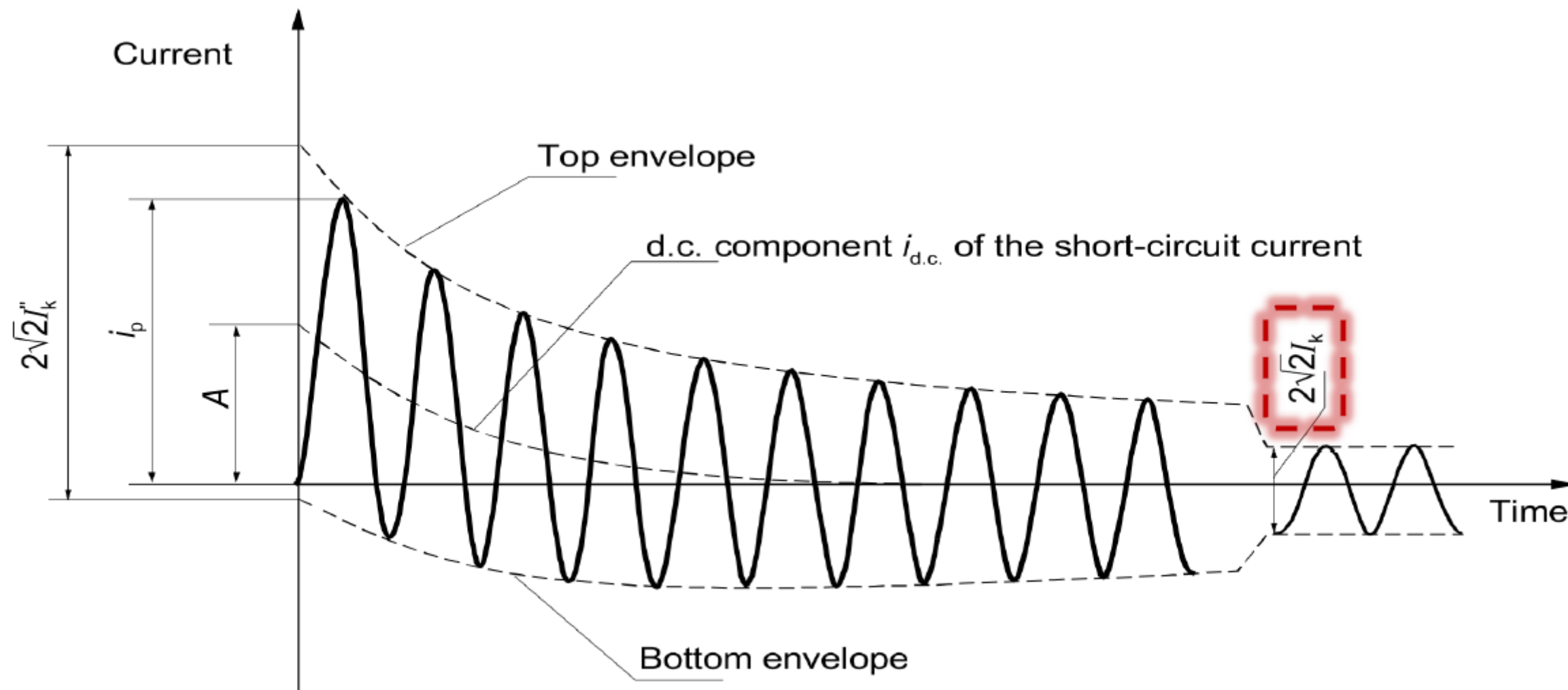
$$R_k/X_k = 0.13/0.12 = 1.08$$

From the diagram $\kappa \approx 1.06$

From the equation:

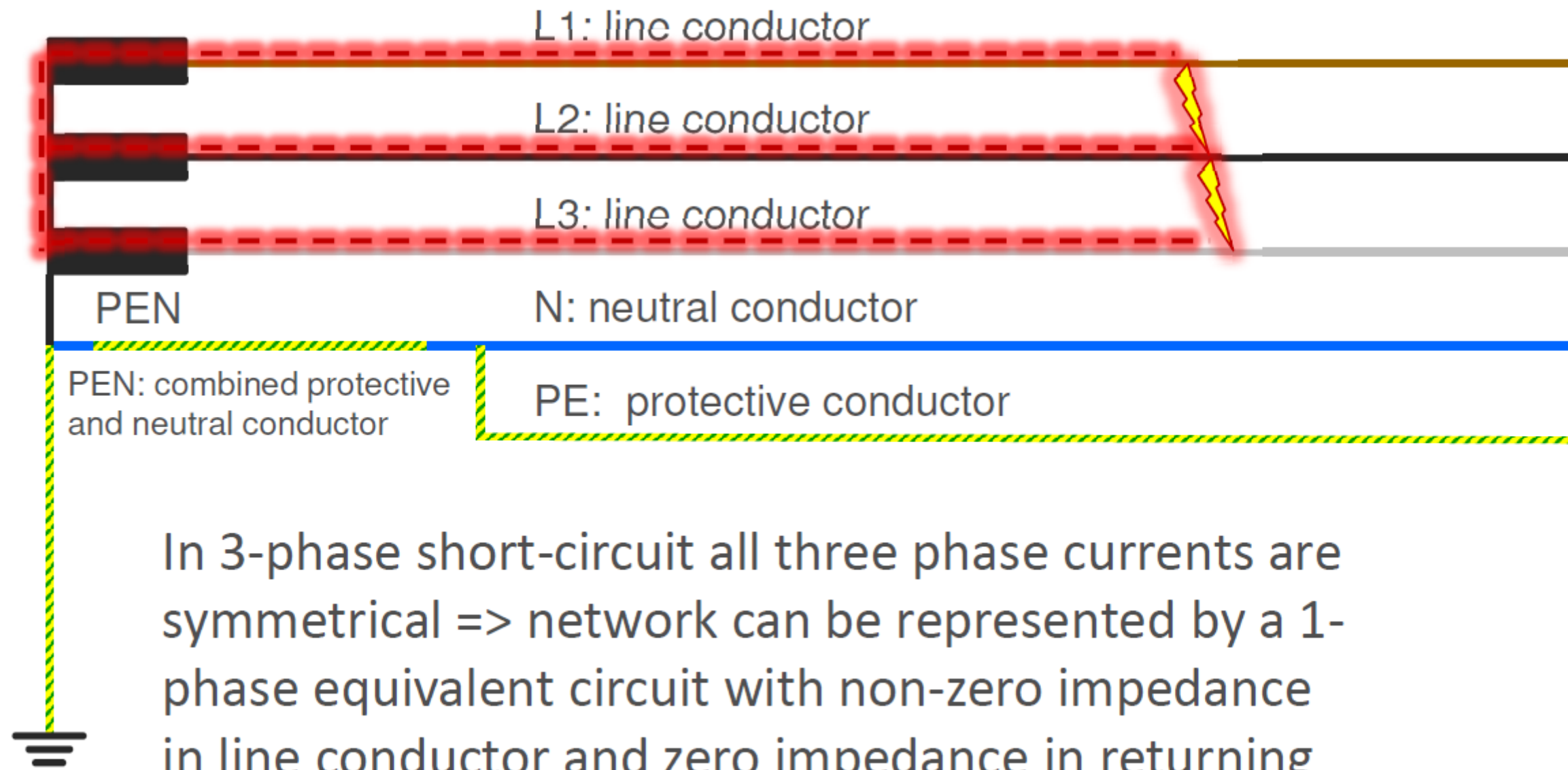
$$\begin{aligned}\kappa &= 1.022 + 0.969 \cdot e^{\frac{-3.03 \cdot R_k}{X_k}} \\ &= 1.022 + 0.969 \cdot e^{\frac{-3.03 \cdot 0.013}{0.012}} = 1.06\end{aligned}$$

Steady-state 3-phase short-circuit current I_k



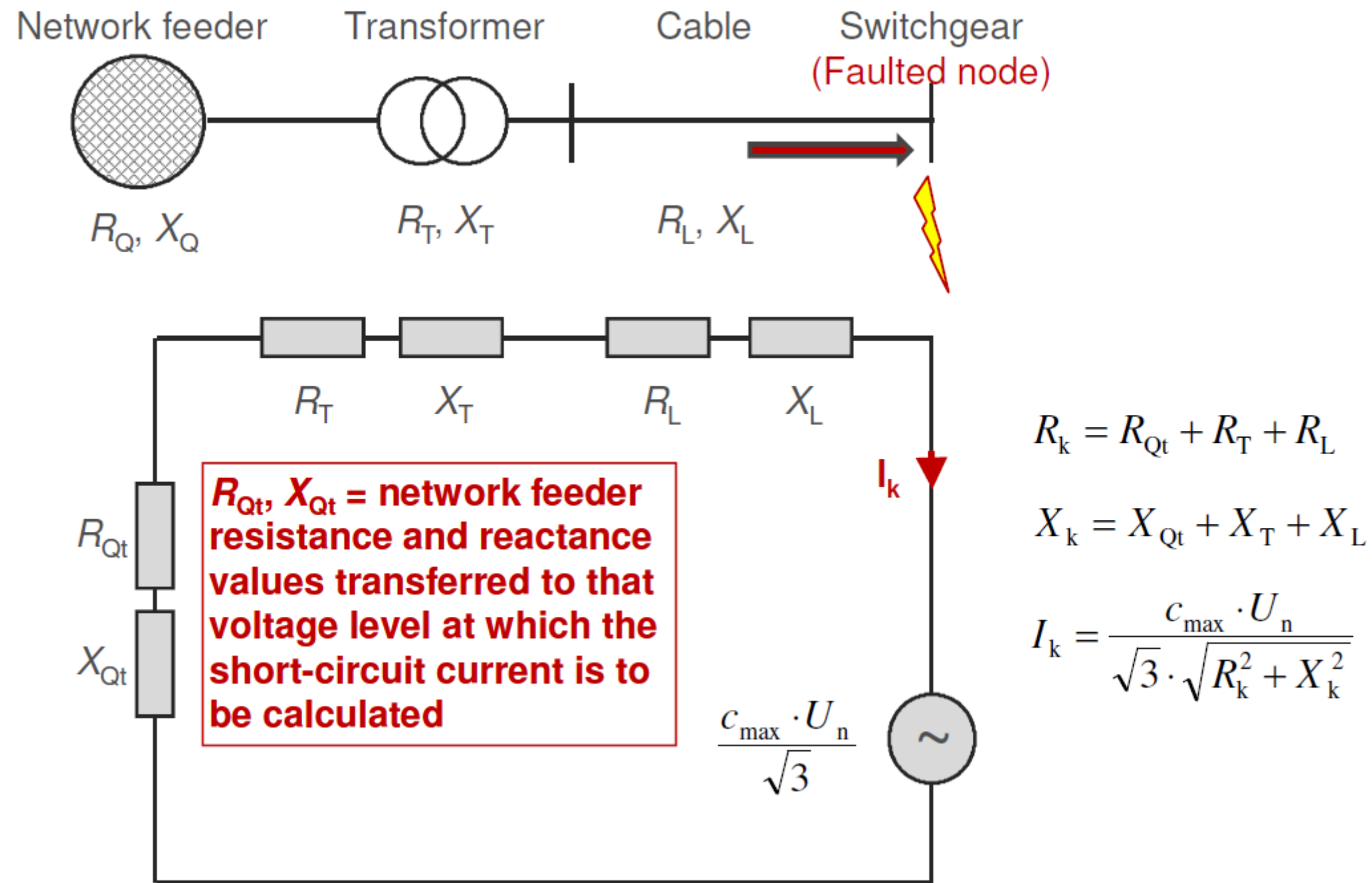
- ◆ Steady-state short-circuit current I_k
 - R.m.s. value of the short-circuit current which remains after the decay of the transient phenomena

Steady-state 3-phase short-circuit current I_k



In 3-phase short-circuit all three phase currents are symmetrical => network can be represented by a 1-phase equivalent circuit with non-zero impedance in line conductor and zero impedance in returning conductor.

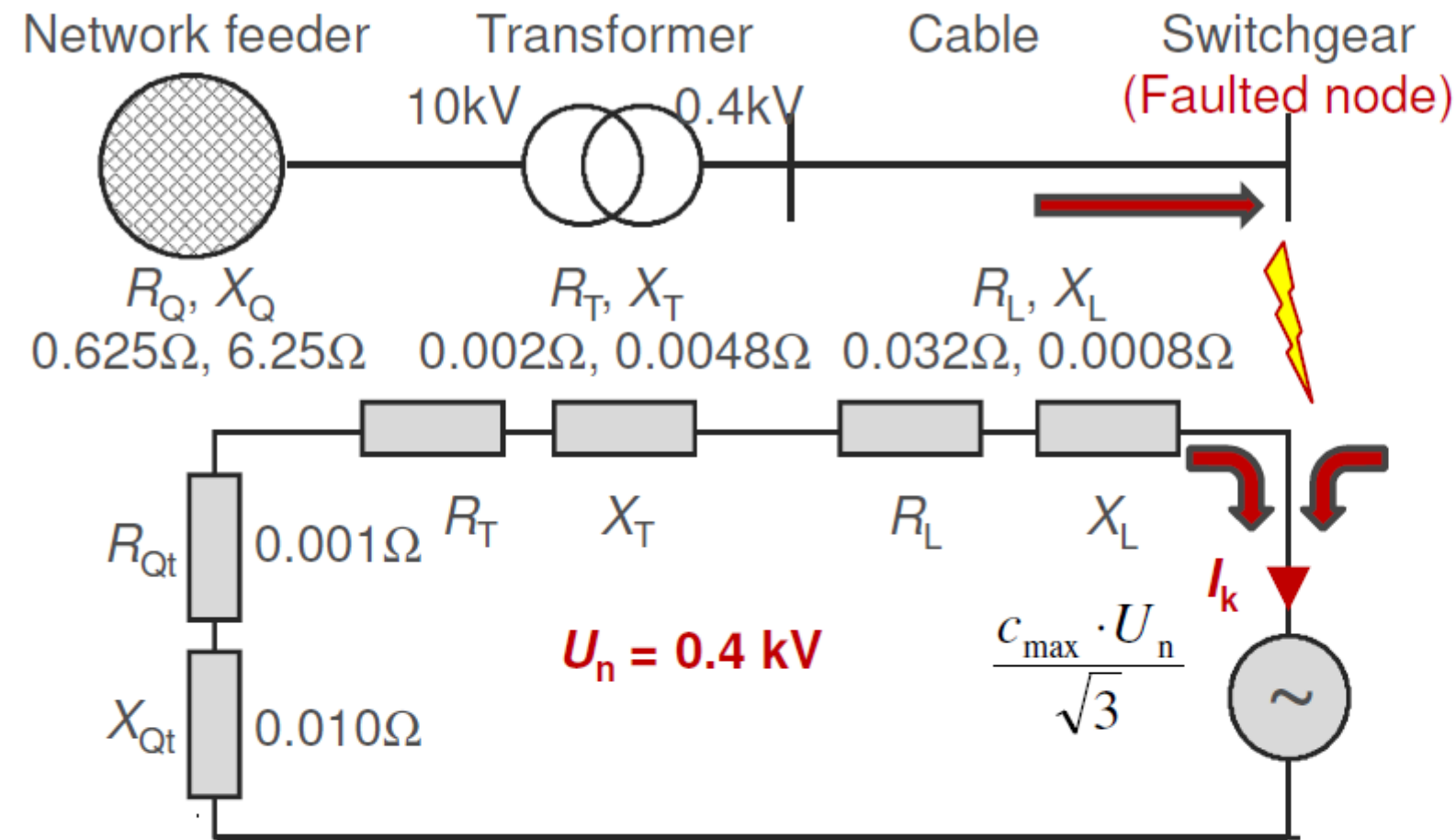
Steady-state 3-phase short-circuit current I_k



Steady-state 3-phase short-circuit current I_k

- Voltage factor $c_{\max}$:
 - $0.1 \text{ kV} < U_n \leq 1 \text{ kV}$ $c_{\max} = 1.05 \dots 1.10$
 - $U_n > 1 \text{ kV}$ $c_{\max} = 1.10$
- When calculating steady-state short-circuit current:
 - Resistance R_L of lines obtained at a temperature of 20°C
 - Fault impedance Z_F assumed to be zero
 - **Asynchronous machines are excluded** (feeding current to the faulted node only at the beginning of the sc)
 - **Synchronous machines are included** (feeding current to the faulted node due to excitation)
 - System configuration and the maximum contribution from power plants/network feeders which lead to the maximum value of short-circuit current at the short-circuit location

Steady-state 3-phase short-circuit current I_k



$$R_k = R_{Qt} + R_T + R_L = 0.001 + 0.002 + 0.032 = 0.035 \Omega$$

$$X_k = X_{Qt} + X_T + X_L = 0.010 + 0.0048 + 0.0008 = 0.0156 \Omega$$

$$I_k = \frac{c_{\max} \cdot U_n}{\sqrt{3} \cdot \sqrt{R_k^2 + X_k^2}} = \frac{1.05 \cdot 400 \text{ V}}{\sqrt{3} \cdot \sqrt{0.035^2 + 0.0156^2}} = 6328 \text{ A} = 6.3 \text{ kA}$$

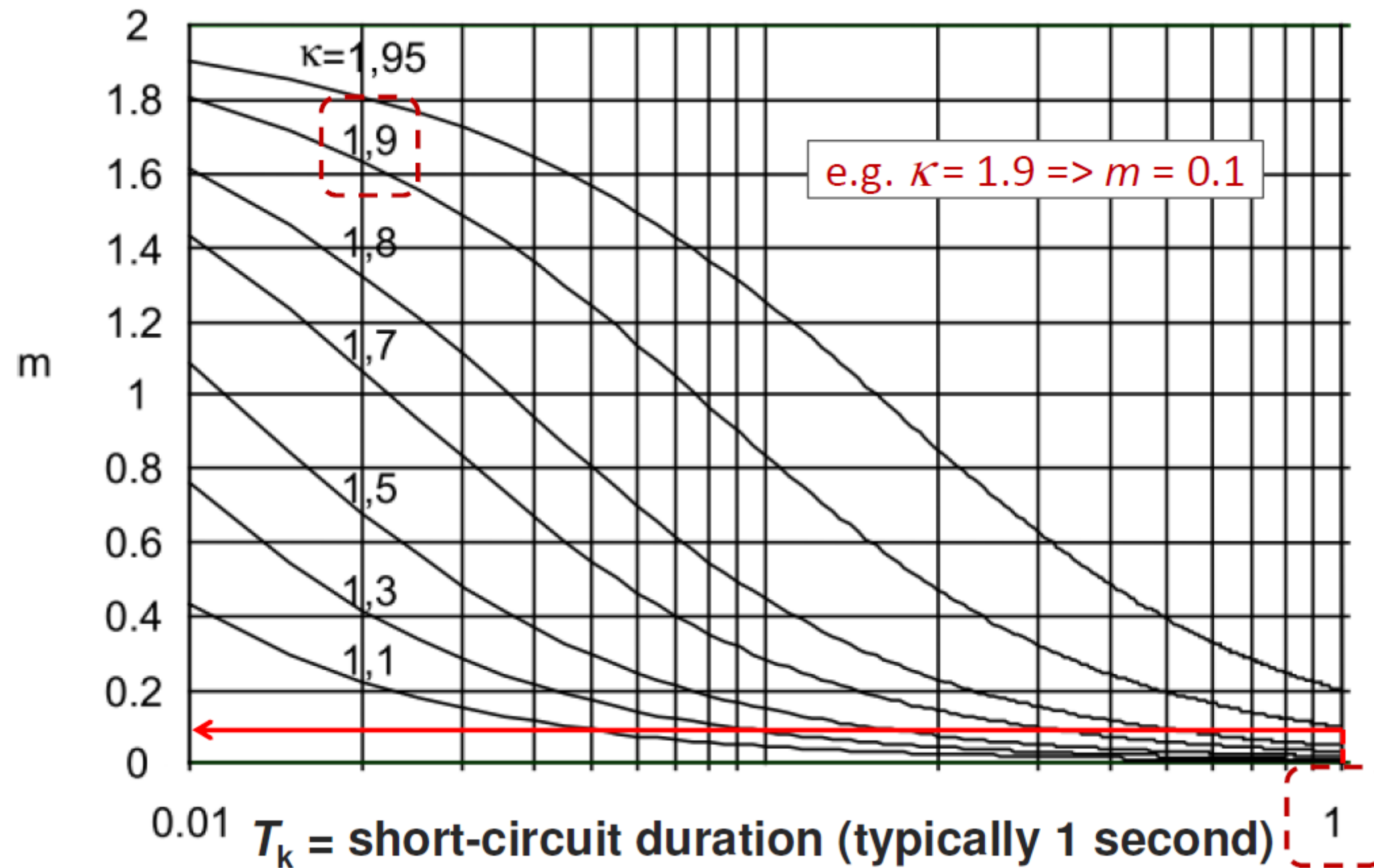
Thermal equivalent short-circuit current I_{th}

- The r.m.s. value of a current having the same thermal effect and the same duration as the actual short-circuit current, which may contain a d.c. component and may subside in time
- Typically calculated for short-circuit duration $T_k = 1 \text{ s}$
- Can be calculated with the following formula

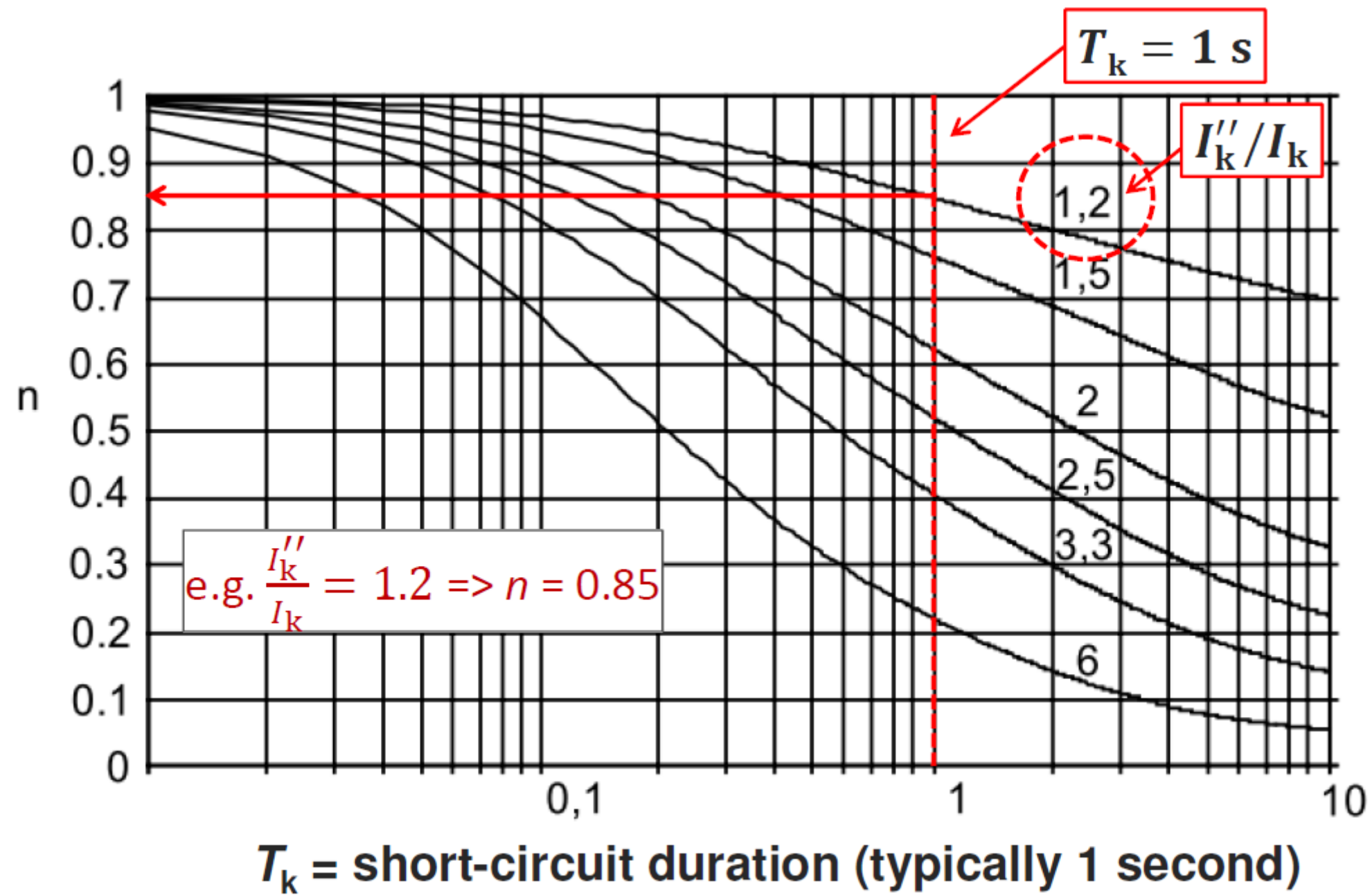
$$I_{th} = I_k'' \cdot \sqrt{m + n}$$

- m and n are time-dependent heat effects of the d.c. and a.c. components of the short-circuit current and can be obtained from figures on next pages

Thermal equivalent short-circuit current I_{th}



Thermal equivalent short-circuit current I_{th}



Thermal equivalent short-circuit current I_{th}

According to previous example:

$$\kappa = 1.06 \Rightarrow m = 0.01$$

$$\frac{I_k'''}{I_k} = \frac{13.7 \text{ kA}}{6.3 \text{ kA}} = 2.2 \Rightarrow n = 0.58$$

$$I_{th} = I_k''' \cdot \sqrt{m + n} = 13.7 \text{ kA} \cdot \sqrt{0.01 + 0.58} = 10.5 \text{ kA}$$

Withstand capacity

- To ensure fault-free function of electrical equipment, it is always required that it is sufficiently rated not only for continuous current but also for thermal and mechanical effects of short-circuit currents
- I_e = rated normal current, current that a switching device in closed position can carry continuously
- I_{cw} = rated short-time withstand current, current that a switching device in closed position can carry during a specified short time (typical value 1 s)
- I_{pk} = rated peak withstand current, capacity against mechanical stress caused by peak sc-current

Withstand capacity

- Rated normal current I_e is selected from standardized values to be higher than the continuous load current

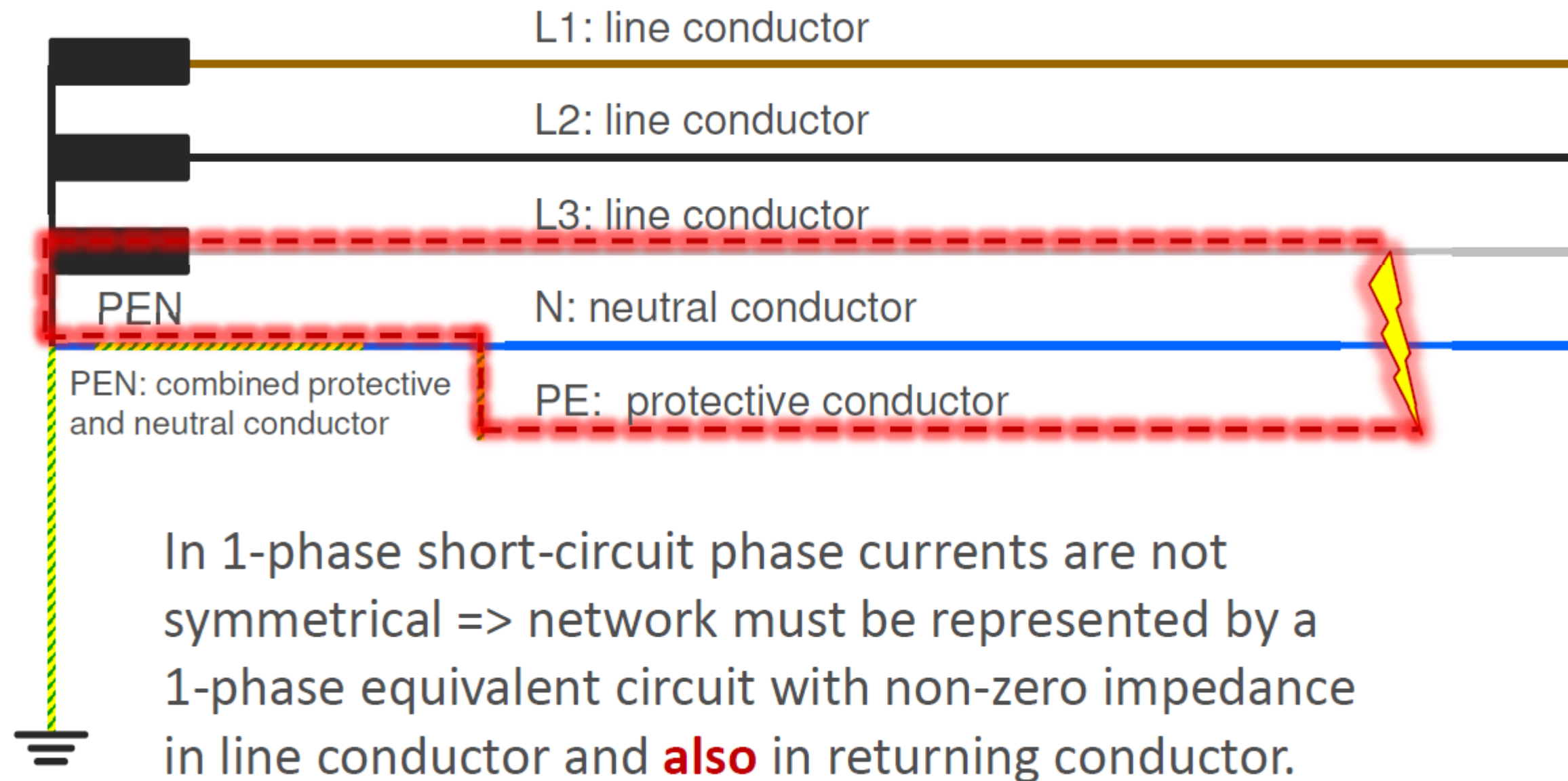
I_e (A)	125	250	400	630	800	1000	1250	1600	2000	2500	3150
--------------	-----	-----	-----	-----	-----	------	------	------	------	------	------

- Rated short-time and peak withstand currents I_{cw} and I_{pk} are selected from standardized values to be higher than the thermal sc-current I_{th} and peak sc-current i_p

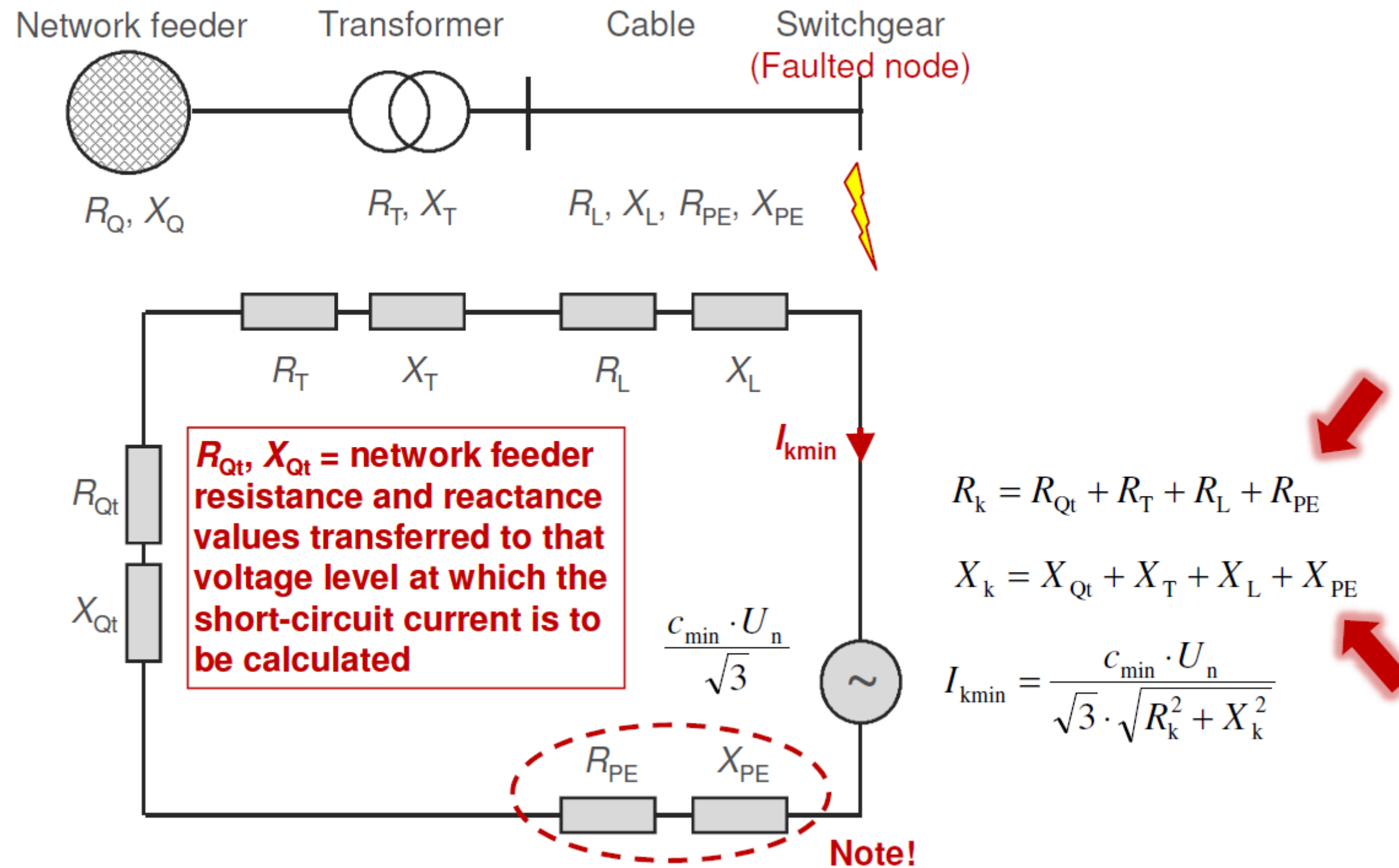
I_{cw} (kA)	5	6.3	8	10	12.5	16	20	25	31.5	40	50	63	80	100
I_{pk} (kA)	12.5	16	20	25	31.5	40	50	63	80	100	125	160	200	250

$$I_{pk} = 2.5 \cdot I_{cw}$$

Minimum 1-phase short-circuit current I_{kmin}



Minimum 1-phase short-circuit current I_{kmin}



Minimum 1-phase short-circuit current I_{kmin}

- ◆ NOTE: This is not according to IEC 60909
- ◆ This is more practical way to calculate minimum 1-phase short-circuit current and leads to a small error compared to IEC 60909 which uses method of symmetrical components

Minimum 1-phase short-circuit current I_{kmin}

- Voltage factor c_{min} :
 - $0.1 \text{ kV} < U_n \leq 1 \text{ kV}$ $c_{min} = 0.95$ (see next page)
 - $U_n > 1 \text{ kV}$ $c_{min} = 1.00$
- When calculating minimum 1-phase short-circuit current:
 - Resistance R_L of lines obtained at a temperature of 80°C
 - Fault impedance Z_F assumed to be zero
 - **Asynchronous machines are excluded** (feeding current to the faulted node only at the beginning of the sc)
 - System configuration and the minimum contribution from power plants/network feeders/**synchronous machines** which lead to the minimum value of short-circuit current at the short-circuit location

Minimum 1-phase short-circuit current I_{kmin}

Nominal voltage U_n	Voltage factor c for the calculation of	
	maximum short-circuit currents $c_{\max}^{1)}$	minimum short-circuit currents $c_{\min}$
Low voltage 100 V to 1 000 V (IEC 60038, table I)	$1,05^{3)}$ $1,10^{4)}$	0,95
Medium voltage >1 kV to 35 kV (IEC 60038, table III)	1,10	1,00
High voltage²⁾ >35 kV (IEC 60038, table IV)		

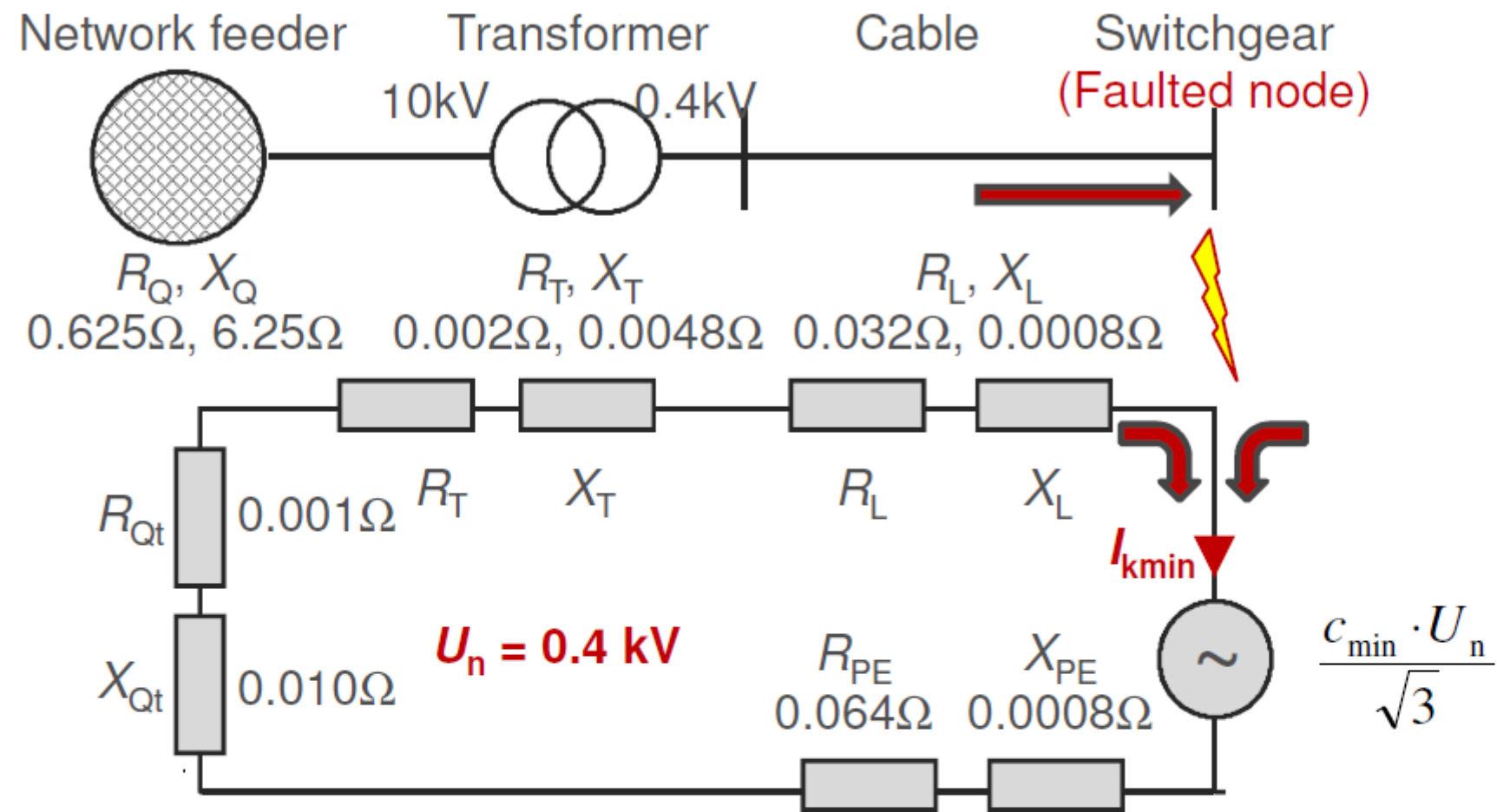
¹⁾ $c_{\max}U_n$ should not exceed the highest voltage U_m for equipment of power systems.

²⁾ If no nominal voltage is defined $c_{\max}U_n = U_m$ or $c_{\min}U_n = 0,90 \times U_m$ should be applied.

³⁾ For low-voltage systems with a tolerance of +6 %, for example systems renamed from 380 V to 400 V.

⁴⁾ For low-voltage systems with a tolerance of +10 %.

Minimum 1-phase short-circuit current I_{kmin}



$$R_k = R_{Qt} + R_T + R_L + R_{PE} = 0.001 + 0.002 + 0.032 + 0.064 = 0.099 \Omega$$

$$X_k = X_{Qt} + X_T + X_L + X_{PE} = 0.010 + 0.0048 + 0.0008 + 0.0008 = 0.0164 \Omega$$

$$I_{kmin} = \frac{c_{min} \cdot U_n}{\sqrt{3} \cdot \sqrt{R_k^2 + X_k^2}} = \frac{0.95 \cdot 400 \text{ V}}{\sqrt{3} \cdot \sqrt{0.099^2 + 0.0164^2}} = 2186 \text{ A} = 2.2 \text{ kA}$$

Use of short-circuit currents

	I_k''	i_p	I_k	I_{th}	I_{kmin}
Electrical equipment short-time withstand current				X	
Electrical equipment peak withstand current		X			
Protection equipment breaking capacity	X				
Protection equipment making capacity		X			
Protection equipment tripping settings	X				X
Discrimination, selectivity	X				X

Exercises



Exercise 1: Equivalent circuits of network components

1. Feeder network nominal voltage is $U_{nQ} = 400$ V, low-voltage system tolerance is 6 % and symmetrical sc-current at the feeder connection point is $I_{kQ}'' = 500$ A. Calculate network feeder impedance Z_Q , resistance R_Q and reactance X_Q .
2. Feeder network system nominal voltage is $U_{nQ} = 400$ V, low-voltage system tolerance is 6 % and short-circuit power at the feeder connection point is $S_{kQ}'' = 1$ MVA. Calculate network feeder impedance Z_Q , resistance R_Q and reactance X_Q .
3. Calculate AMCMK 3x35/16 cable **line conductor** resistance R_L and reactance X_L . Cable length is $l = 120$ m, frequency is $f = 50$ Hz and conductor temperature is $\theta_e = 70$ °C. Part of the datasheet is shown below.
4. Calculate AMCMK 3x35/16 cable **PE-conductor** resistance R_L and reactance X_L . Cable length is $l = 120$ m, frequency is $f = 50$ Hz and conductor temperature is $\theta_e = 70$ °C. Part of the datasheet is shown below.

Exercise 1: Equivalent circuits of network components (ctd.)

5. Transformer is connected to network with system nominal voltage of $U_n = 10$ kV. Rated apparent power of the transformer is $S_{rT} = 1250$ kVA and rated transformation ratio is $U_{1r}/U_{2r} = 10/0.4$ kV. Relative resistive component of the short-circuit voltage is $u_{Rr} = 1$ % and short-circuit voltage at rated current is $u_{kr} = 6$ %. Calculate ohmic value for short-circuit impedance Z_T , resistance R_T and reactance X_T using transformer secondary voltage.
6. Transfer the former problem Z_T to the transformer primary voltage level (10 kV).
7. Group of low-voltage motors are connected to $U_n = 0.4$ kV network. Sum of the apparent powers of the motors in a group is $S_{rM} = 1.0$ MVA. Calculate Z_M , X_M and R_M .

Exercise 1: Equivalent circuits of network components (ctd.)

PRODUCT NAME			AMCMK 3x16/10 1 kV	AMCMK 3x25/16 1 kV	AMCMK 3x35/16 AN 1 kV	AMCMK 3x50/16 AN 1 kV	AMCMK 3x70/21 AN 1 kV	AMCMK 3x95/29 AN 1 kV	AMCMK 3x120/41 AN 1 kV
STK-code			0622157	0622158	0622159	0622160	0622161	0622162	0622163
ELECTRICAL DATA (2)									
Maximum DC resistance of phase conductor.	conductor 20°C	Ω/km	1,91	1,20	0,868	0,641	0,443	0,320	0,253
AC resistance of phase and neutral conductor (1)	conductor 70°C	Ω/km	2,3	1,4	1,0	0,77	0,53	0,39	0,31
Maximum DC resistance of PE conductor	conductor 20°C	Ω/km	1,83	1,15	1,15	1,15	0,868	0,641	0,443
Inductance (1)		mH/km	0,28	0,28	0,26	0,25	0,24	0,24	0,23

References



References

Slides reproduced from

- ♦ P. Puttonen (2016): Lecture on Short-Circuits, part of *Energy Transmission* lecture series, 8th-10th November 2016, FH Vorarlberg, Dornbirn